

Master Thesis

Facilitating the Integration of Borehole Thermal Energy
Storage in District Energy Systems: Improving the Fidelity of
Design Tools

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Abstract

Borehole Thermal Energy Storage (BTES) systems offer a promising solution for increasing the flexibility and efficiency of integrated district heating and cooling systems. However, representing their dynamic operation and interaction with connected heat pumps and chillers within techno-economic optimization frameworks results in computationally demanding optimization problems. It therefore remains important to identify which modeling assumptions are essential for accurate system design and which can be simplified without substantially affecting the optimal solution.

This thesis develops and evaluates a techno-economic framework for the simultaneous sizing and operation of BTES-integrated heating and cooling systems. Building upon an existing mixed-integer nonlinear programming (MINLP) formulation, three operational strategies are investigated to assess how different thermodynamic representations influence optimal technology sizing and annual system cost. The reference MINLP is subsequently compared with a mixed-integer linear programming (MILP) reformulation based on piecewise McCormick relaxations and with a steady-state heuristic sizing approach. In addition, the influence of linearization, charging-period optimization, modeling assumptions, and system complexity is systematically assessed.

Comparison with a detailed Modelica simulation showed that the optimization model accurately reproduces the seasonal evolution of the operating temperatures, with an annual mean absolute error of 2.51 K and a root-mean-square error of 2.84 K. Among the operational strategies, allowing the condenser and evaporator temperatures to vary throughout the year reduced the annual system cost by approximately 2.7% without introducing a noticeable computational penalty. The original three-segment MILP required approximately 2.6 times longer solution times than the corresponding MINLP while increasing the annual cost by 6.1%. Further analysis showed that this discrepancy originated not only from the McCormick approximation itself but also from the interaction between the relaxation and the discrete BTES design space. Refining the linearization demonstrated that increasing the number of McCormick segments does not necessarily produce a solution closer to the nonlinear optimum. In both case studies, the one-segment MILP reproduced the MINLP solution more closely than the original three-segment formulation while requiring substantially lower computational effort.

The simplification study showed that reducing the McCormick discretization and pre-

scribing the seasonal charging period provided significant computational savings while maintaining close agreement with the MINLP solution. In contrast, fixing the operating temperatures produced the largest reduction in runtime but also the largest deviation from the nonlinear reference solution. Finally, applying the methodology to a more technologically diverse energy system demonstrated that, although the MINLP became computationally more demanding, the one-segment fixed-period MILP solved the larger optimization problem with a higher speedup compared to the MINLP than the simpler EMPA case study while maintaining close agreement with the reference non-linear model. This result indicates that increasing system flexibility can improve the computational performance of simplified MILP formulations.

Overall, the results demonstrate that computational performance depends not only on model size but also on thermodynamic representation, linearization strategy, and system flexibility. The proposed framework provides practical guidance for selecting an appropriate level of modeling fidelity according to the objectives of BTES system design, showing that a one-segment MILP offers an effective compromise between computational efficiency and solution accuracy, while the steady-state heuristic remains a valuable tool for rapid preliminary sizing.

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Use of Artificial Intelligence Tools

Artificial intelligence (AI)-based tools were used during the preparation of this thesis to support the writing process. In particular, ChatGPT [1] was used for:

- improving the grammar, clarity, and readability of the thesis
- refining the structure and flow of selected sections
- assisting with the preparation and formatting of tables and figures

The use of AI was limited to supporting the writing, presentation, and formatting of the thesis. All scientific content, including the research objectives, methodology, modeling approach, analysis, interpretation of the results, and conclusions, was developed and critically reviewed by the author, who assumes full responsibility for the content of this thesis.

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List of Abbreviations

Acronyms

ω Capital Recovery Factor

BTES Borehole Thermal Energy Storage

CAPEX Capital Expenditure

COP Coefficient of Performance

EMPA Swiss Federal Laboratories for Materials Science and Technology

EU European Union

GHX Ground Heat Exchanger

MES Multi Energy System

MILP Mixed-Integer Linear Programming

MINLP Mixed-Integer Nonlinear Programming

OPEX Operational Expenditure

List of Mathematical Symbols

Latin symbols

Symbol	Description	Unit
A	Heat transfer area	m^2
A_{top}	BTES top surface area	m^2
c	Unit energy price	CHF/kWh
c_v	Volumetric heat capacity	$\text{kWh m}^{-3}\text{K}^{-1}$
C	Thermal capacity	kWh K^{-1}
D	Borehole depth	m
d	Storage diameter	m
E	Thermal energy	kWh
G	Natural gas input	kW
i	Interest rate	–
J	Annualized cost	CHF/yr
k	Ground thermal conductivity	$\text{W m}^{-1}\text{K}^{-1}$
n	Number of boreholes / lifetime	–
P	Electrical power	kW
Q	Thermal power	kW
S	Installed capacity	kW
T	Temperature	$^{\circ}\text{C}$
U	Heat transfer coefficient	$\text{W m}^{-2}\text{K}^{-1}$
UA	Overall heat transfer conductance	W K^{-1}
V	Storage volume	m^3
x	Continuous decision variable	–
y	Binary decision variable	–
z	Auxiliary variable	–

Greek symbols

Symbol	Description	Unit
Δt	Simulation timestep	h
ΔT	Temperature difference	K
δ	BTES design variable	–
η	Efficiency	–
λ	Specific investment cost	CHF/kW
ω	Capital recovery factor	–

Indices and subscripts

Symbol	Description
t	Time step
i	Technology
j	BTES design option
s	Piecewise-linear segment
air	Ambient air
boiler	Boiler
ch	Charging / Chiller
cond	Condenser
cool	Cooling
dem	Demand
dis	Discharging
evap	Evaporator
heat	Heating
hp	Heat pump
loss	Thermal loss
sol	Solar thermal
top	Top insulation
L	Lower bound
U	Upper bound

1 Introduction

The European Union (EU) has set the objective of achieving climate neutrality by 2050 [2]. Reaching this target requires a substantial reduction in greenhouse gas emissions across all sectors of the economy and a transition towards more sustainable energy systems [3]. To support this transition, significant investments have been made in renewable energy technologies and energy efficiency measures [4, 5, 6]. As a result, the share of renewable energy sources in Europe reached 25.2% in 2024, with projections indicating an increase to 42.5% by 2030 [7].

In parallel, the EU has introduced several policy mechanisms aimed at accelerating decarbonization [8, 9]. One of the most important is the Emission Trading System (ETS), which places a cost on greenhouse gas emissions and provides an economic incentive for industries to reduce their carbon footprint and invest in low-carbon technologies [10]. Such measures contribute to the broader objective of carbon neutrality by encouraging emission reductions across different sectors of the economy [11].

A significant portion of decarbonization efforts has focused on the electricity sector, where renewable generation technologies such as wind and solar power have experienced rapid growth [12, 13, 14]. However, electricity represents only part of the overall energy system. Approximately 50% of the total final energy consumption in Europe is associated with heating and cooling applications, and nearly 70% of this demand is still supplied by fossil fuels [15]. Consequently, the decarbonization of the heating and cooling sector remains one of the key challenges in achieving climate neutrality.

Among the technologies that can contribute to this transition is the BTES, which has attracted increasing attention [16, 17]. BTES systems utilize the ground as a large-scale thermal storage medium, injecting and extracting heat through borehole heat exchangers [18, 19]. By exploiting the high thermal capacity of the underground, BTES systems enable seasonal thermal energy storage, allowing surplus heat generated during one period of the year to be utilized at a later time. This capability can improve overall system efficiency, reduce reliance on fossil fuels, and increase the flexibility of district energy systems [20].

A notable example of BTES implementation is the Swiss Federal Laboratories for Materials Science and Technology (EMPA) campus, where seasonal thermal storage is used to recover excess heat during summer and utilize it during winter. The integration of BTES

with heat pumps and chillers allows waste heat to be recovered and reused while supporting the electrification of heating and cooling services [21]. Such systems demonstrate the potential of seasonal thermal storage to improve the performance of multi-energy systems and contribute to the decarbonization of the built environment.

1.1 Literature Gap and Contribution

The growing interest in BTES technology has led to a wide range of studies investigating its design, operation, and integration within energy systems. Several publications provide an overview of underground thermal energy storage technologies and discuss the principles governing the operation and performance of BTES systems [22]. Other studies focus specifically on the design of borehole heat exchanger fields, seeking to improve storage performance through the optimization of borefield layouts and thermal interactions between boreholes [23].

In addition to borefield-level optimization, BTES has been investigated within the broader context of district and Multi Energy System (MES) [24]. These studies typically consider the simultaneous sizing and operation of BTES together with surrounding technologies such as heat pumps, chillers, boilers, and thermal storage systems. A notable example is the work of Fiorentini et al. [25], who developed an optimization framework for the design of district heating and cooling systems incorporating seasonal thermal storage while minimizing both capital and operational expenditures. Similar system-level optimization approaches have been proposed for the design of multi-energy systems with seasonal storage technologies [26, 27].

Although considerable progress has been made in the optimization of BTES-integrated energy systems, relatively limited attention has been given to the influence of model fidelity on the resulting design decisions. Existing studies often employ simplified thermal representations in order to maintain computational tractability, while more detailed BTES models have primarily been developed for simulation purposes [28, 29]. Consequently, there remains a lack of understanding regarding the extent to which modeling assumptions influence optimization results and whether the additional complexity associated with higher-fidelity models is justified.

More specifically, it remains unclear which modeling assumptions have the greatest impact on technology sizing, system operation, and overall economic performance. Furthermore,

the relative importance of thermodynamic representation, mathematical formulation, and system complexity has not yet been systematically quantified within a common modeling framework.

1.2 Research Idea

The objective of this thesis is to investigate how different modeling assumptions and levels of modeling fidelity influence the optimal design, operation, and computational performance of BTES-integrated energy systems. Rather than focusing exclusively on improving model accuracy, the work aims to understand the trade-offs between modeling complexity, computational efficiency, and the quality of the resulting design decisions.

To achieve this objective, the thesis investigates two complementary aspects that influence optimization-based system design.

The first aspect concerns the operational assumptions adopted for the BTES-connected heat pump and chiller. Three alternative operational strategies are considered, ranging from simplified scalar decision variables to formulations in which all variables vary across the entire optimization horizon. The objective is to quantify how these different operational assumptions influence technology sizing, system operation, annual system cost, and computational performance.

The second aspect concerns the mathematical formulation of the optimization problem. Alongside the Mixed-Integer Nonlinear Programming (MINLP) formulation proposed by Fiorentini et al. [25], a Mixed-Integer Linear Programming (MILP) reformulation and a steady-state heuristic simulation model are developed. Although all three approaches represent the same physical system, they differ considerably in their mathematical formulation, computational complexity, and solution methodology. This enables the influence of different levels of modeling fidelity on the optimal solution and computational performance to be systematically assessed.

The proposed framework is evaluated using two case studies of increasing system complexity. This enables the investigation of how the structure of the energy system influences computational performance and solution quality, as well as how the introduction of additional technologies affects the dependence of the optimization on the nonlinear BTES subsystem.

Together, these investigations provide a comprehensive framework for evaluating the in-

fluence of both operational assumptions and modeling fidelity on the design of BTES-integrated energy systems. The different approaches are compared in terms of technology sizing, operational behavior, economic performance, and computational effort. Through this analysis, the thesis identifies the modeling assumptions that have the greatest influence on the optimization results.

2 Methodology

The methodology used to investigate how model fidelity affects the predicted design and operation of a MES with a BTES system is presented in this chapter. The work builds on the optimization-oriented framework proposed by Fiorentini et al. [25], but the objective is not to reproduce the original model alone. Instead, the original framework is used as a reference and extended through different thermodynamic representations, mathematical formulations, and case study configurations.

2.1 Methodological Framework

Figure 1 summarizes the methodological workflow followed throughout this thesis. After defining the case study, the methodology is divided into two complementary branches.

The first branch focuses on the sizing approaches. Three models are developed and compared: a steady-state heuristic model, a mixed-integer linear programming (MILP) formulation, and a mixed-integer nonlinear programming (MINLP) formulation. The heuristic model is compared with the mathematical formulations to assess its suitability as a rapid preliminary sizing tool, while the MILP formulation is used to evaluate the influence of different modeling features through successive simplifications. The MINLP formulation serves as the reference optimization model and is used to investigate the effect of different operational strategies on the optimal system design.

The second branch focuses on model verification. A high-fidelity Modelica model is used to verify the thermal behavior of the reference MINLP formulation before the operational strategy analysis is performed.

Together, these analyses provide a systematic comparison of the different modeling approaches in terms of technology sizing, annual system cost, BTES operation and computational performance.

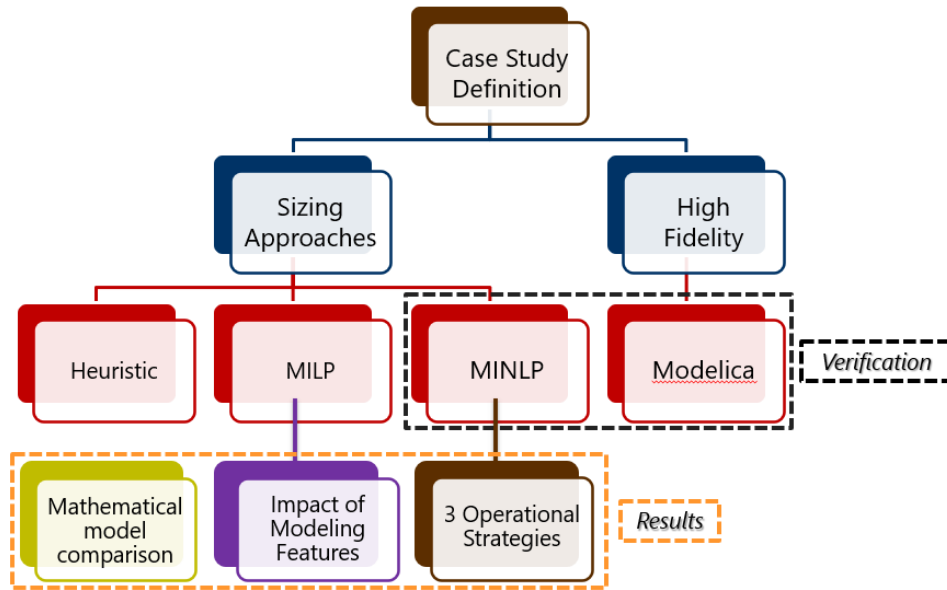


Figure 1: Methodological framework

System Description

The system considered in this work is a district heating and cooling plant with seasonal thermal storage. The plant must satisfy a time-dependent heating demand and a time-dependent cooling demand over a yearly horizon.

The set of potential candidates to be introduced to the system consists of a heat pump connected to the BTES, a chiller connected to the BTES, air-source heating and cooling technologies, a boiler, a cooling tower, solar thermal collectors, an electricity grid connection, and a natural gas grid connection. The complete system configuration is shown in Figure 2

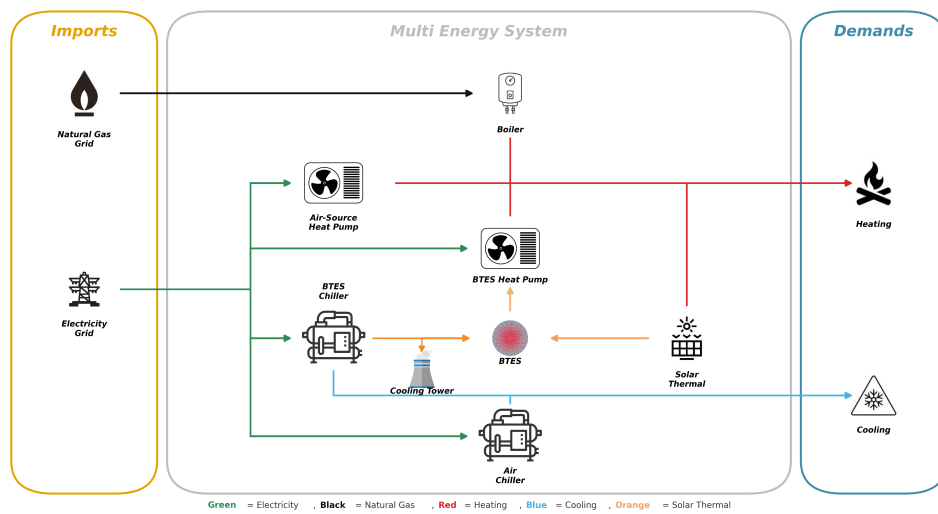


Figure 2: Borehole thermal energy storage system connected into a multi energy system

For the purpose of this study, both the extended case study shown in Figure 2 and the EMPA campus case study shown in Figure 3 are considered, with particular emphasis on the EMPA case study. The EMPA case represents a simpler energy system designed to supply the heating and cooling demands of a small campus. The system incorporates a BTES connected to a heat pump and a chiller. The cooling demand can only be supplied by the chiller, whereas the heating demand can be met by either the boiler or the BTES-connected heat pump.

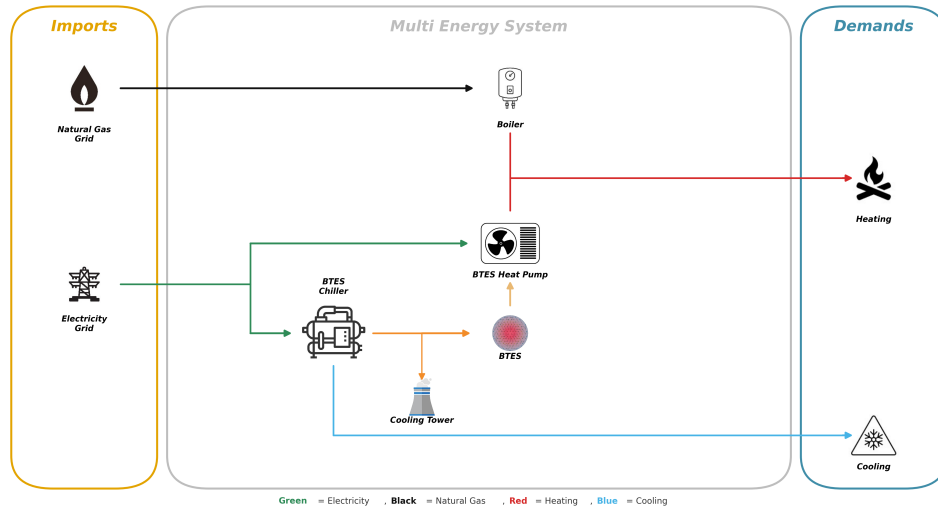


Figure 3: Borehole thermal energy storage system connected into a multi energy system for the EMPA campus

All the parameters used for the modeling of the system can be found in Appendix A1.

Objective Function

The objective of the optimization model is to identify the combination of technology capacities and operating schedules that minimizes the annual cost of supplying the heating and cooling demands. Since both investment decisions and operational decisions influence the overall economics of the system, the optimization simultaneously determines technology sizing and system operation over the simulation horizon.

The total annual system cost is expressed as the sum of annualized Capital Expenditure (CAPEX) and annual Operational Expenditure (OPEX),

$$\min J_{\text{tot}} = J_{\text{capex}} + J_{\text{opex}}, \quad (1)$$

The operating cost is associated with the consumption of external energy carriers and is

calculated as

$$J_{\text{opex}} = J_{\text{el}} + J_{\text{gas}}, \quad (2)$$

The annual electricity cost and annual gas cost are calculated using Eqs. (3) and (4) respectively.

$$J_{\text{el}} = \sum_{t \in \mathcal{T}} c_{\text{el},t} P_{\text{grid},t} \Delta t, \quad (3)$$

$$J_{\text{gas}} = \sum_{t \in \mathcal{T}} c_{\text{gas},t} G_{\text{grid},t} \Delta t, \quad (4)$$

where $c_{\text{el},t}$ and $c_{\text{gas},t}$ are the cost of electricity and natural gas at time t , respectively. $P_{\text{grid},t}$ and $G_{\text{grid},t}$ are the electricity and gas input at time t , respectively.

The annualized capital cost accounts for the installed capacities of all technologies included in the system,

$$J_{\text{capex}} = \sum_{i \in \mathcal{I}} J_{\text{capex},i} \quad (5)$$

where each term represents the annualized investment cost of the corresponding technology i .

For a generic technology i , the annualized investment cost is calculated as

$$J_i = \omega_i \lambda_i S_i, \quad (6)$$

with λ being the specific investment cost, S being the installed capacity and the Capital Recovery Factor (ω) of the technology i being calculated with equation 7 [30]

$$\omega_i = \frac{i(1+i)^{n_i}}{(1+i)^{n_i} - 1} \quad (7)$$

Unlike the remaining technologies, the BTES cost depends on the selected borefield design. The annualized BTES investment cost is therefore calculated from the total drilling length, and the costs related to the Ground Heat Exchanger (GHX) [31]

$$J_{\text{BTES}} = \omega_{\text{BTES}} \lambda_{\text{GHX}} \sum_{j \in \mathcal{J}} \delta_j D_{\text{BTES},j} n_{\text{BTES},j}, \quad (8)$$

with D being the borehole depth and n the number of boreholes of the BTES option j . δ is a decision variable that is 1 for any BTES design and 0 for the absence of the BTES. The objective function, therefore captures the fundamental trade-off explored throughout this thesis. Larger technology capacities generally increase the annualized investment cost but may reduce operating costs through improved efficiency or increased utilization of stored thermal energy.

Heating and Cooling Balances

Before describing the behavior of the BTES, it is necessary to define the thermal demands that the system must satisfy.

The district heating demand can be supplied by multiple technologies depending on the case study configuration. These include the BTES-connected heat pump, the air-source heat pump, the gas boiler and the solar thermal collectors. The corresponding heating and cooling balances are formulated as [32, 33, 34]

$$Q_{\text{dem,h},t} = \sum_{i \in \mathcal{I}_{\text{heat}}} Q_{i,t} \quad \forall t \in \mathcal{T}. \quad (9)$$

$$Q_{\text{dem,c},t} = \sum_{i \in \mathcal{I}_{\text{cool}}} Q_{i,t}, \quad \forall t \in \mathcal{T}, \quad (10)$$

where $Q_{\text{dem,h},t}$ is the heating demand at time t and $Q_{\text{dem,c},t}$ is the cooling demand at time t that are to be supplied by the set of heating technologies $i \in \mathcal{I}_{\text{heat}}$ and cooling technologies $i \in \mathcal{I}_{\text{cool}}$ respectively.

The thermal output of each technology i is constrained by its installed capacity S_i .

$$0 \leq Q_{i,t} \leq S_i, \quad \forall t \in \mathcal{T}, \quad (11)$$

The boiler converts natural gas into useful thermal energy with an efficiency η_{boiler} according to

$$Q_{\text{boiler},t} = \eta_{\text{boiler}} G_{\text{boiler},t}, \quad \forall t \in \mathcal{T}, \quad (12)$$

The operation of electricity and gas-driven technologies contributes to the total electrical and gas demand of the energy system. The imported electricity $P_{\text{grid},t}$ and the imported

gas $G_{\text{grid},t}$ are given by Eqs. (13) and (14) respectively.

$$P_{\text{grid},t} = \sum_{i \in \mathcal{I}_{\text{el}}} P_{i,t}, \quad \forall t \in \mathcal{T}. \quad (13)$$

$$G_{\text{grid},t} = G_{\text{boiler},t}, \quad \forall t \in \mathcal{T}, \quad (14)$$

where $\mathcal{I}_{\text{el}}$ denotes the set of electrically driven technologies

BTES Model

Having established the system-level energy balances, the next step is to describe the behavior of the BTES. The BTES enables the temporal shifting of thermal energy between periods of cooling demand and periods of heating demand. Consequently, the evolution of the BTES temperature directly affects both the amount of energy that can be stored and the efficiency of the heat pump and chiller connected to the storage.

The BTES is represented using a lumped-capacitance approach. Instead of modeling the temperature distribution inside the storage volume, the BTES is characterized by a single average temperature.

The storage is selected from a predefined set of BTES designs. Each design is characterized by its volume, depth, borefield conductance, and number of boreholes. A binary variable δ_j is assigned to each design option $j \in \mathcal{J}$,

$$\delta_j = \begin{cases} 1, & \text{if BTES design } j \text{ is selected} \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

with the constraint

$$\sum_{j \in \mathcal{J}} \delta_j = 1. \quad (16)$$

The available BTES design options are indexed by $j \in \mathcal{J}$. Each BTES option is described by its volume $V_{\text{BTES},j}$, depth $D_{\text{BTES},j}$, borefield conductance $UA_{\text{BTES},j}$, and number of boreholes $n_{\text{GHX},j}$, as shown in table 1.

Table 1: BTES design options from Fiorentini et al.

Option	Diameter (m)	Depth (m)	Volume (m ³)	UA (kW/K)	Boreholes
1	53.4	53.4	119 200	22.5	158
2	61.2	61.2	179 600	33.8	207
3	70.0	70.0	269 400	50.6	271
4	80.1	80.1	404 100	76.0	355
5	91.7	91.7	606 100	113.9	466
6	–	–	–	–	–

This formulation ensures that exactly one BTES configuration is selected during the optimization. Option 6 represents the absence of seasonal storage, allowing the optimization to determine whether the BTES is economically beneficial.

The amount of thermal energy that can be stored inside the BTES depends on the thermal properties of the surrounding ground and the selected storage volume.

The thermal capacity of the storage is calculated as

$$C_{\text{BTES},j} = c_{v,g} V_{\text{BTES},j}, \quad (17)$$

where $c_{v,g}$ is the volumetric heat capacity of the ground.

The state variable used to represent the storage is the average BTES temperature $T_{\text{BTES},t}$. The temperature evolution is governed by an energy balance between heat entering the storage $Q_{\text{BTES},\text{ch},t}$, heat extracted from the storage $Q_{\text{BTES},\text{dis},t}$, and heat losses to the environment $Q_{\text{loss},t}$,

$$T_{\text{BTES},t+1} = T_{\text{BTES},t} + \frac{\Delta t}{C_{\text{BTES}}} (Q_{\text{BTES},\text{ch},t} - Q_{\text{BTES},\text{dis},t} - Q_{\text{loss},t}), \quad (18)$$

The charging term consists of heat rejected from the chiller condenser and solar thermal energy directed to the storage,

$$Q_{\text{BTES},\text{ch},t} = Q_{\text{BTES},\text{chiller},t} + Q_{\text{BTES},\text{solar},t}. \quad (19)$$

The discharging term corresponds to the heat extracted by the BTES-connected heat pump evaporator,

$$Q_{\text{BTES},\text{dis},t} = Q_{\text{evap},\text{hp},t}. \quad (20)$$

The total BTES heat loss consists of two components,

$$Q_{\text{loss},t} = Q_{\text{loss,top},t} + Q_{\text{loss,ground},t}. \quad (21)$$

The first component represents heat transfer through the top insulation layer,

$$Q_{\text{loss,top},t} = UA_{\text{top},j} (T_{\text{BTES},t} - T_{\text{air},t}), \quad (22)$$

where $UA_{\text{top},j}$ is the equivalent heat-transfer conductance of the top insulation layer for BTES design j , calculated as

$$UA_{\text{top},j} = U_{\text{top}} A_{\text{top},j}, \quad (23)$$

with the top layer area being calculated as

$$A_{\text{top},j} = \pi \left(\frac{d_j}{2} \right)^2, \quad (24)$$

where d is the diameter of the top layer of the BTES.

The second component represents conductive heat transfer between the storage and the surrounding ground,

$$Q_{\text{loss,ground},t} = UA_{\text{ground},j} (T_{\text{BTES},t} - T_g), \quad (25)$$

The equivalent ground-loss conductance is calculated as

$$UA_{\text{ground},j} = k_g \frac{D_j}{2} h_{\text{lf}}, \quad (26)$$

Positive values of Eqs. (22) and (25) correspond to heat losses from the BTES to its surroundings, while negative values represent heat gains when the ambient air or surrounding ground temperature exceeds the storage temperature.

For the BTES system, a cyclic operation throughout the year was assumed, with this condition ensuring that the storage finishes the year at the same state in which it started. Without this constraint, the optimization could artificially improve the objective by ending the horizon with an overcharged or depleted storage.

$$T_{\text{BT},|\mathcal{T}|} = T_{\text{BT},0}. \quad (27)$$

The exchange of heat between the BTES and the connected heat pump or chiller is limited by the borefield heat-transfer conductance.

For charging,

$$Q_{\text{BTES,chiller},t} \leq U A_{\text{BTES},j} \Delta T_{\text{ch},t}, \quad (28)$$

while for discharging,

$$Q_{\text{BTES,dis},t} \leq U A_{\text{BTES},j} \Delta T_{\text{hp},t}. \quad (29)$$

These constraints ensure that the heat exchanged with the storage remains physically feasible and directly link the BTES operation to the thermodynamic representation of the heat pump and chiller.

Charging and Discharging Periods

In practical operation, a BTES system follows a seasonal operating strategy in which thermal energy is accumulated during periods of excess heat availability and extracted during periods of heating demand. Allowing the storage to charge and discharge simultaneously would lead to physically unrealistic operating schedules and artificial energy circulation within the system. To ensure a more realistic representation of BTES operation, a binary variable is introduced to distinguish between charging and discharging periods.

The charging period is represented by the binary variable

$$\delta_t^{\text{ch}} \in \{0, 1\}, \quad \forall t \in \mathcal{T}, \quad (30)$$

where $\delta_t^{\text{ch}} = 1$ indicates that the BTES operates in charging mode, while $\delta_t^{\text{ch}} = 0$ corresponds to the discharging period.

The charging period is not predetermined. Instead, its start and end are selected by the optimization model within predefined seasonal windows corresponding to spring and autumn, respectively. Consequently, the duration of the charging season is optimized together with the remaining system design and operational variables.

To enforce the selected operating mode, indicator constraints are introduced. During the charging period, heat extraction from the BTES is prohibited,

$$\delta_t^{\text{ch}} = 1 \Rightarrow Q_{\text{dis},t} = 0, \quad \forall t \in \mathcal{T}, \quad (31)$$

whereas during the discharging period, all charging flows to the BTES are forced to zero,

$$\delta_t^{\text{ch}} = 0 \Rightarrow Q_{\text{ch,chiller},t} = 0, \quad \forall t \in \mathcal{T}, \quad (32)$$

$$\delta_t^{\text{ch}} = 0 \Rightarrow Q_{\text{ch,solar},t} = 0, \quad \forall t \in \mathcal{T}. \quad (33)$$

This formulation ensures that the BTES can only operate in one mode at each timestep while allowing idle operation when no thermal energy is exchanged. Compared with a fixed seasonal schedule, optimizing the charging period provides additional operational flexibility and allows the model to determine the most economically favorable charging and discharging windows. As this assumption modifies the operational behavior of the storage without altering its physical representation, the charging-period optimization is treated as a modeling feature and its influence on BTES sizing, system operation, and computational performance is assessed in section 3.

2.2 Operational Strategy Options

Following the development of the BTES model, three operational strategies are introduced to describe the thermodynamic interaction between the BTES and the connected heat pump and chiller. These strategies represent different approaches for operating the thermal storage during system operation. The objective is to investigate how the selected operational strategy influences the optimal design of the integrated energy system, identify the strategy that minimizes the total annualized cost, and evaluate how different operational choices affect the sizing of the system components.

For all strategies, the performance of the BTES-connected heat pump and chiller is represented using the inverse Coefficient of Performance (COP), which provides a linear relationship between the operating temperature and the equipment performance [25]:

$$COP_{\text{hp,BTES},t}^{-1} = a_{\text{hp}} T_{\text{evap},t} + b_{\text{hp}}, \quad (34)$$

$$COP_{ch,BTES,t}^{-1} = a_{ch}T_{cond,t} + b_{ch} \quad (35)$$

where a and b are operational parameters presented in Appendix A1, while T_{cond} and T_{evap} are the condenser and evaporator temperatures respectively.

The three strategies represent alternative approaches for controlling the interaction between the BTES and the connected heat pump and chiller. As illustrated in Figure 4, their operation is described using four main parameters: the heat-pump evaporator temperature, $T_{evap,t}$, the chiller condenser temperature, $T_{cond,t}$, the temperature difference between the BTES and the heat-pump evaporator, $\Delta T_{hp,t}$, and the temperature difference between the chiller condenser and the BTES, $\Delta T_{ch,t}$. The selected operational strategy determines which of these quantities are kept constant and which are allowed to vary over time. The remaining operating temperatures are then obtained from their relationship with the time-dependent BTES temperature, $T_{BTES,t}$, which is determined by the storage energy balance presented in Section 2.1. All operating temperatures and temperature differences are constrained within predefined limits to ensure feasible system operation.

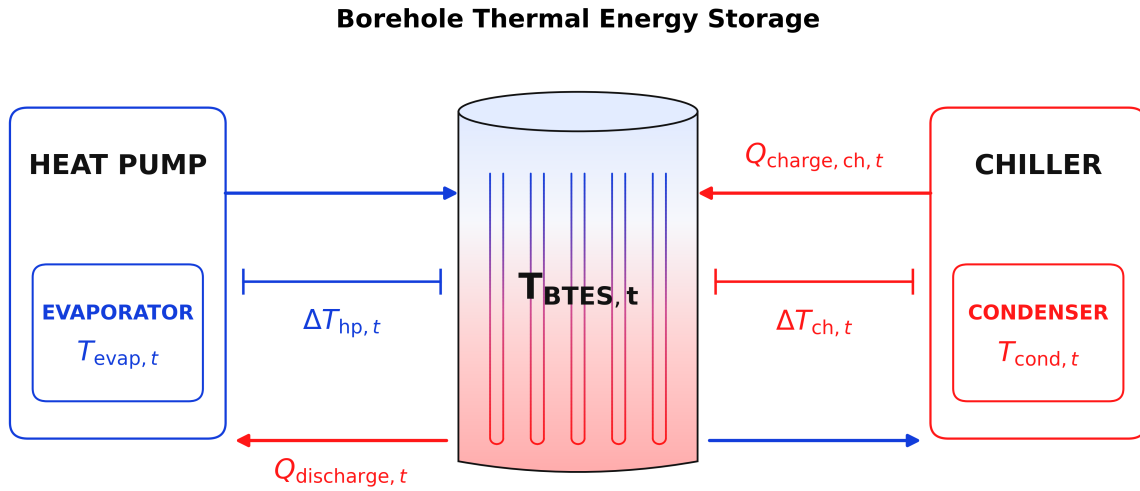


Figure 4: Operating temperatures and temperature differences governing the interaction of the BTES with the heat-pump evaporator during discharge and the chiller condenser during charging. Where T_{BTES} is the average BTES temperature of the system.

Strategy I: Constant Temperature Difference

The first strategy assumes constant temperature differences between the BTES and the connected heat pump and chiller throughout the simulation horizon. Consequently, the

optimization determines two scalar decision variables, ΔT_{hp} and ΔT_{ch} , while the evaporator and condenser temperatures are obtained from

$$T_{\text{evap},t} = T_{\text{BTES},t} - \Delta T_{\text{hp}}, \quad \forall t \in \mathcal{T}, \quad (36)$$

$$T_{\text{cond},t} = T_{\text{BTES},t} + \Delta T_{\text{ch}}, \quad \forall t \in \mathcal{T}. \quad (37)$$

Since the BTES temperature varies over time, the evaporator and condenser temperatures follow the storage temperature, while the temperature differences remain constant. The resulting operating temperatures are then substituted into Eqs. (34) and (35) to determine the corresponding heat pump and chiller performance.

Strategy II: Constant Evaporator and Condenser Temperatures

In the second strategy, the evaporator and condenser temperatures are selected as the scalar decision variables and remain constant throughout the simulation horizon. Consequently, the corresponding temperature differences vary according to the BTES temperature and are obtained from Eqs. (36) and (37).

The optimized evaporator and condenser temperatures are subsequently used to evaluate the heat pump and chiller performance through Eqs. (34) and (35). Since both temperatures remain constant, the corresponding coefficients of performance are also constant over the entire simulation period. The BTES temperature, however, continues to evolve according to the storage energy balance, resulting in time-varying temperature differences between the storage and the connected devices.

Strategy III: Dynamic Evaporator and Condenser Temperatures

The third strategy extends Strategies I and II by allowing the evaporator and condenser temperatures to vary throughout the simulation horizon. Instead of optimizing a single annual operating temperature, the optimization determines the evaporator and condenser temperatures at every timestep.

The corresponding temperature differences are obtained from the BTES temperature through Eqs. (36) and (37).

The heat pump and chiller performance is then evaluated using Eqs. (34) and (35), re-

sulting in time-dependent coefficients of performance that adapt to the selected operating temperatures at each timestep.

Compared to Strategy II, this formulation increases the operational flexibility of the heat pump and chiller by allowing their operating temperatures to vary over time while maintaining the same BTES representation. This additional flexibility increases the number of decision variables and may therefore increase computational effort.

2.3 Mathematical Formulations

The operational strategies presented in the previous section define how the heat pump and chiller are represented physically. However, the same physical system can be formulated mathematically in different ways. These formulations may differ significantly in terms of computational effort, solution quality and scalability, even when they describe the same physical system.

The second fidelity dimension investigated in this thesis therefore concerns the mathematical representation of the problem. Three formulations are considered: a MINLP formulation, a MILP reformulation, and a steady-state heuristic simulation model.

The MINLP formulation retains the nonlinear relationships between equipment performance and thermodynamic state variables. The MILP formulation approximates these nonlinear relationships through piecewise-linear representations and McCormick relaxations. Finally, the steady-state heuristic removes the optimization problem entirely and instead estimates system performance through engineering calculations and seasonal energy balances.

The purpose of this comparison is not to determine which formulation is universally superior. Rather, it is to investigate how much fidelity is lost when computationally simpler formulations are adopted and whether these simplifications lead to different design and operational conclusions.

MINLP Formulation

The MINLP formulation represents the highest-fidelity optimization approach considered in this thesis. The model preserves the nonlinear relationships between thermodynamic temperatures, equipment efficiencies and electrical power consumption, allowing the interaction between the BTES and the connected technologies to be represented directly

within the optimization problem.

The principal source of nonlinearity originates from the relationship between thermal output, coefficient of performance and electrical consumption. For any electrically driven technology, the electrical power consumption is given by

$$P_{i,t} = Q_{i,t} \cdot COP_{i,t}^{-1}. \quad (38)$$

For the BTES-connected heat pump and the BTES-connected chiller this yields

$$P_{\text{hp,BTES},t} = a_{\text{hp}} Q_{\text{hp,BTES},t} T_{\text{evap},t} + b_{\text{hp}} Q_{\text{hp,BTES},t}, \quad (39)$$

$$P_{\text{ch,BTES},t} = a_{\text{ch}} Q_{\text{ch,BTES},t} T_{\text{cond},t} + b_{\text{ch}} Q_{\text{ch,BTES},t}. \quad (40)$$

The products $Q_{\text{hp,BTES},t} T_{\text{evap},t}$ and $Q_{\text{ch,BTES},t} T_{\text{cond},t}$ contain two optimization variables and therefore create bilinear non-convex terms. These terms are responsible for the nonlinear nature of the optimization problem.

Figure 5 illustrates the resulting bilinear relationship for the BTES-connected heat pump at three fixed evaporator temperatures. For a fixed temperature, electrical consumption varies linearly with the thermal output. However, the slope changes with the evaporator temperature. Since both $Q_{\text{hp,BTES},t}$ and $T_{\text{evap},t}$ are optimization variables in the MINLP, the complete relationship cannot be represented by a single linear function. Instead, it forms a bilinear surface over the joint operating domain.

The resulting optimization problem contains both binary and nonlinear constraints. Consequently, the MINLP provides the closest mathematical representation of the underlying physical system but also exhibits the highest computational complexity among the formulations considered in this thesis.

MILP Reformulation

This subsection explains how the nonlinear terms of the MINLP are reformulated into an MILP problem using a piecewise McCormick approximation [35].

The motivation for developing the MILP formulation is computational tractability. The MILP aims to retain the main structure of the physical model while replacing the bilinear products with linear constraints.

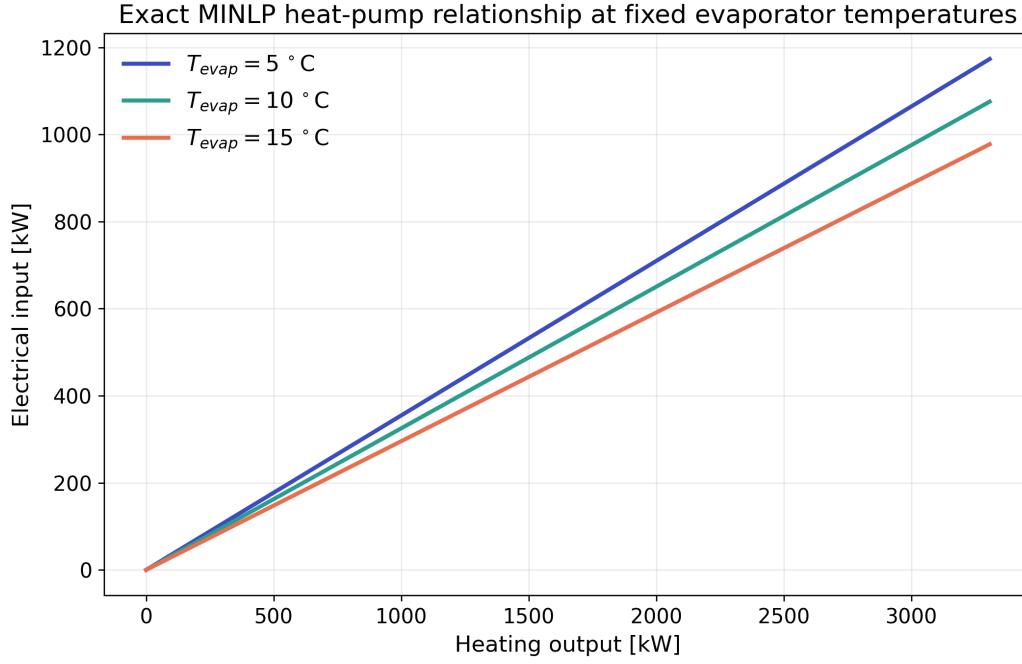


Figure 5: Electrical consumption of the BTES-connected heat pump as a function of its thermal output at evaporator temperatures of 5, 10, and 15 °C

The piecewise McCormick approximation is selected because the relevant nonlinearities are products of two continuous decision variables, namely thermal power and temperature. A direct piecewise-affine representation would require partitioning both variables and would substantially increase the size of the annual hourly model. The selected approach instead partitions only the temperature range, providing a computationally manageable formulation whose accuracy can be improved by increasing the number of segments. Starting from the heat pump power equation, the nonlinear part is the product between heat pump output and evaporator temperature. To isolate this product, an auxiliary variable is introduced,

$$z_{hp,t} = Q_{hp,BTES,t} T_{evap,t}, \quad \forall t \in \mathcal{T}. \quad (41)$$

The heat pump electrical consumption can then be written in linear form as

$$P_{hp,BTES,t} = a_{hp} z_{hp,t} + b_{hp} Q_{hp,BTES,t}, \quad \forall t \in \mathcal{T}. \quad (42)$$

The same reformulation is applied to the chiller. Starting from

$$P_{ch,BTES,t} = a_{ch} z_{ch,t} + b_{ch} Q_{ch,BTES,t}, \quad \forall t \in \mathcal{T}. \quad (43)$$

The remaining task is to approximate the auxiliary product variables $z_{\text{hp},t}$ and $z_{\text{ch},t}$ using linear constraints. This is done by dividing the temperature domain into a finite number of segments. For the heat pump, the evaporator temperature range is divided into segments $s \in \mathcal{S}_{\text{hp}}$, each with lower and upper bounds $T_{\text{evap},s}^L$ and $T_{\text{evap},s}^U$.

A binary variable $y_{\text{hp},t,s}$ indicates whether segment s is active at timestep t ,

$$y_{\text{hp},t,s} = \begin{cases} 1, & \text{if evaporator temperature segment } s \text{ is active at timestep } t, \\ 0, & \text{otherwise.} \end{cases} \quad (44)$$

Exactly one segment must be selected at each timestep,

$$\sum_{s \in \mathcal{S}_{\text{hp}}} y_{\text{hp},t,s} = 1, \quad \forall t \in \mathcal{T}. \quad (45)$$

The evaporator temperature is then disaggregated by segment,

$$T_{\text{evap},t} = \sum_{s \in \mathcal{S}_{\text{hp}}} T_{\text{evap},t,s}, \quad \forall t \in \mathcal{T}, \quad (46)$$

where $T_{\text{evap},t,s}$ is the portion of the evaporator temperature assigned to segment s . The segment bounds are enforced through

$$T_{\text{evap},s}^L y_{\text{hp},t,s} \leq T_{\text{evap},t,s} \leq T_{\text{evap},s}^U y_{\text{hp},t,s}, \quad \forall t \in \mathcal{T}, \forall s \in \mathcal{S}_{\text{hp}}. \quad (47)$$

The heat pump thermal output is also disaggregated by segment,

$$Q_{\text{hp,BTES},t} = \sum_{s \in \mathcal{S}_{\text{hp}}} Q_{\text{hp},t,s}, \quad \forall t \in \mathcal{T}, \quad (48)$$

where $Q_{\text{hp},t,s}$ is the heat pump output assigned to segment s . The segment activation is enforced with

$$0 \leq Q_{\text{hp},t,s} \leq \bar{Q}_{\text{hp},t} y_{\text{hp},t,s}, \quad \forall t \in \mathcal{T}, \forall s \in \mathcal{S}_{\text{hp}}, \quad (49)$$

For each active segment, a local auxiliary product variable is introduced,

$$z_{\text{hp},t,s} \approx Q_{\text{hp},t,s} T_{\text{evap},t,s}. \quad (50)$$

The total heat pump product is obtained by summing over all segments,

$$z_{\text{hp},t} = \sum_{s \in \mathcal{S}_{\text{hp}}} z_{\text{hp},t,s}, \quad \forall t \in \mathcal{T}. \quad (51)$$

The bilinear relation in Eq. (50) is approximated using McCormick envelope constraints. For each timestep and segment, the following four linear inequalities are imposed:

$$z_{\text{hp},t,s} \geq T_{\text{evap},s}^L Q_{\text{hp},t,s}, \quad \forall t, s, \quad (52)$$

$$z_{\text{hp},t,s} \geq \bar{Q}_{\text{hp},t} T_{\text{evap},t,s} + T_{\text{evap},s}^U Q_{\text{hp},t,s} - \bar{Q}_{\text{hp},t} T_{\text{evap},s}^U y_{\text{hp},t,s}, \quad \forall t, s, \quad (53)$$

$$z_{\text{hp},t,s} \leq \bar{Q}_{\text{hp},t} T_{\text{evap},t,s} + T_{\text{evap},s}^L Q_{\text{hp},t,s} - \bar{Q}_{\text{hp},t} T_{\text{evap},s}^L y_{\text{hp},t,s}, \quad \forall t, s, \quad (54)$$

$$z_{\text{hp},t,s} \leq T_{\text{evap},s}^U Q_{\text{hp},t,s}, \quad \forall t, s. \quad (55)$$

Because the temperature range is divided into smaller intervals, the relaxation becomes tighter than a single global McCormick envelope over the full temperature range.

The same procedure is applied to the BTES-connected chiller. The condenser temperature range is divided into segments $s \in \mathcal{S}_{\text{ch}}$, with lower and upper bounds $T_{\text{cond},s}^L$ and $T_{\text{cond},s}^U$. A binary variable $y_{\text{ch},t,s}$ selects the active condenser-temperature segment, while also the same McCormick envelope structure is then imposed for $z_{\text{ch},t,s}$ using $T_{\text{cond},s}^L$, $T_{\text{cond},s}^U$ and $\bar{Q}_{\text{ch},t}$.

This reformulation preserves the main thermodynamic dependence of equipment performance on operating temperature, while allowing the problem to be solved with linear mixed-integer optimization methods.

Figure 6 illustrates the McCormick relaxation at $T_{\text{evap}} = 10^\circ\text{C}$ using one, two, and three temperature segments. The shaded area represents the range permitted by the linear relaxation, while the dashed line shows the exact MINLP relationship. The envelope becomes narrower as the local temperature interval is reduced. However, the accuracy at a specific temperature is not determined only by the number of segments, but also by the position of that temperature within the selected interval. In this example, 10°C is a boundary of the two-segment partition, causing the McCormick bounds to coincide

with the exact relationship. With three segments, the same temperature lies inside the interval from 5 to 15 °C, and a small relaxation gap remains. Thus, increasing the number of segments generally improves the approximation over the complete temperature range, but does not necessarily produce a tighter relaxation at every individual temperature.

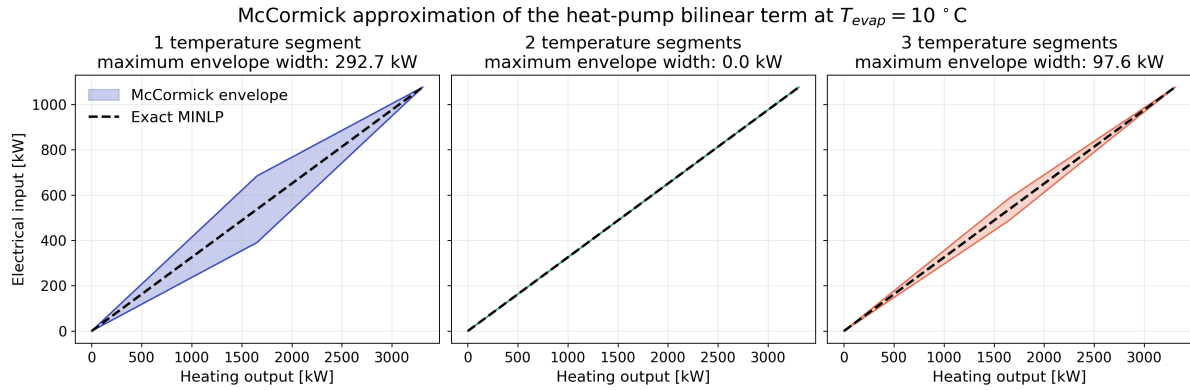


Figure 6: McCormick relaxation using one, two, and three segments for the approximation of the BTES heat-pump bilinear term at $T_{\text{evap}} = 10^\circ\text{C}$

The trade-off of the method is that its accuracy depends on the number of temperature segments, the tightness of the variable bounds, and the operating region selected by the optimizer. For the purpose of this thesis, three segments for both the evaporator and the condenser temperatures are used as the starting point.

Steady-State Heuristic Sizing

To complement the optimization-based formulations, a steady-state heuristic model was developed to provide a computationally inexpensive approximation of the system sizing problem. Rather than solving an optimization problem, the heuristic estimates the capacities of the main technologies through a sequence of engineering calculations based on aggregated seasonal energy balances and simplified thermodynamic assumptions. Similar aggregation approaches have previously been employed for preliminary district energy system sizing [36, 37].

The methodology begins by aggregating the hourly heating and cooling demand profiles into seasonal energy requirements. These energy balances are then used to estimate the amount of thermal energy that can be stored in the BTES during the charging season and the amount that can be recovered during the heating season. The storage requirement is subsequently compared against the available BTES design options, and the smallest feasible configuration is selected.

The capacities of the BTES heat pump and charging chiller are estimated directly from the building peak loads using predefined engineering assumptions, avoiding the need for an optimization routine. Once the equipment has been sized, the annual operation of the selected BTES is simulated using a lumped daily energy balance that accounts for charging, discharging, and thermal losses to both the ground and the ambient environment. The storage temperature is updated sequentially throughout the year while respecting the prescribed operating temperature limits. Any excess charging energy is considered curtailed, whereas insufficient stored energy during the heating season is recorded as unmet thermal demand.

This procedure provides a rapid estimate of the system performance while preserving the dominant physical behavior governing seasonal thermal storage. Although the heuristic cannot determine globally optimal solutions, it offers a useful benchmark for assessing the computational benefits and accuracy of the optimization-based formulations presented in this work.

The complete mathematical formulation of the heuristic model, including the governing energy balances, equipment sizing equations, storage selection procedure, and daily BTES simulation, is provided in Appendix A2.

3 Results

3.1 Verification

For verification, the condenser and evaporator temperatures predicted by the constant ΔT strategy using the MINLP were compared with the corresponding results from the detailed Modelica simulation.

Figure 7 presents the annual operating-temperature trajectories and their hourly differences. During charging, the condenser temperature is compared because the chiller rejects heat to the storage. During discharging, the evaporator temperature is compared because the heat pump extracts heat from the storage.

Both models reproduce the seasonal development of the relevant operating temperatures. The MINLP trajectories are considerably smoother, whereas the Modelica results exhibit short-term fluctuations caused by transient heat transfer, fluid transport, and changes in component loading. During charging, the MINLP predicts a mean condenser temperature of 29.926 °C, compared with 31.315 °C in Modelica. During discharging, the mean evaporator temperatures are 10.868 °C and 9.936 °C, respectively.

The statistical comparison is summarized in Table 2. The mean error during charging is −1.389 K, with a mean absolute error of 2.141 K and an RMSE of 2.406 K. During discharging, the mean error is 0.932 K, while the mean absolute error and RMSE increase to 2.885 K and 3.217 K, respectively. Across both periods, the annual mean error is −0.232 K, the mean absolute error is 2.512 K, and the RMSE is 2.840 K. The maximum absolute hourly error is 8.213 K.

Figure 8 shows the corresponding error distributions, defined as MINLP minus Modelica. The boxplots separate the condenser error during charging from the evaporator error during discharging, while the histograms show the frequency of the paired hourly errors. The dashed vertical lines indicate the mean error of each operating period. The charging errors are predominantly negative, whereas the discharging errors have a positive mean and a wider distribution.

The remaining differences are also related to the simplified representation of component efficiency. Around nominal operating conditions, the optimization model tends to overestimate the COP. Toward the end of the discharging period, the equipment operates more frequently at partial load. These conditions are represented explicitly in Modelica but are

not fully captured by the MINLP performance relationships, resulting in an underestimation of the COP under partial-load operation. The wider evaporator-temperature error distribution during discharging is consistent with this limitation.

Overall, the comparison supports the use of the MINLP for annual system-design optimization and seasonal operating analysis. Nevertheless, the model is not expected to reproduce the short-term component dynamics and partial-load behavior represented by the detailed Modelica simulation.

Table 2: Statistical comparison of condenser temperatures during charging and evaporator temperatures during discharging. Errors are defined as MINLP minus Modelica.

Quantity	Charging: condenser	Discharging: evaporator
Mean MINLP temperature [°C]	29.926	10.868
Mean Modelica temperature [°C]	31.315	9.936
Mean bias error [K]	-1.389	0.932
Median error [K]	-2.307	1.796
Mean absolute error [K]	2.141	2.885
Root-mean-square error [K]	2.406	3.217
Maximum absolute error [K]	3.543	8.213

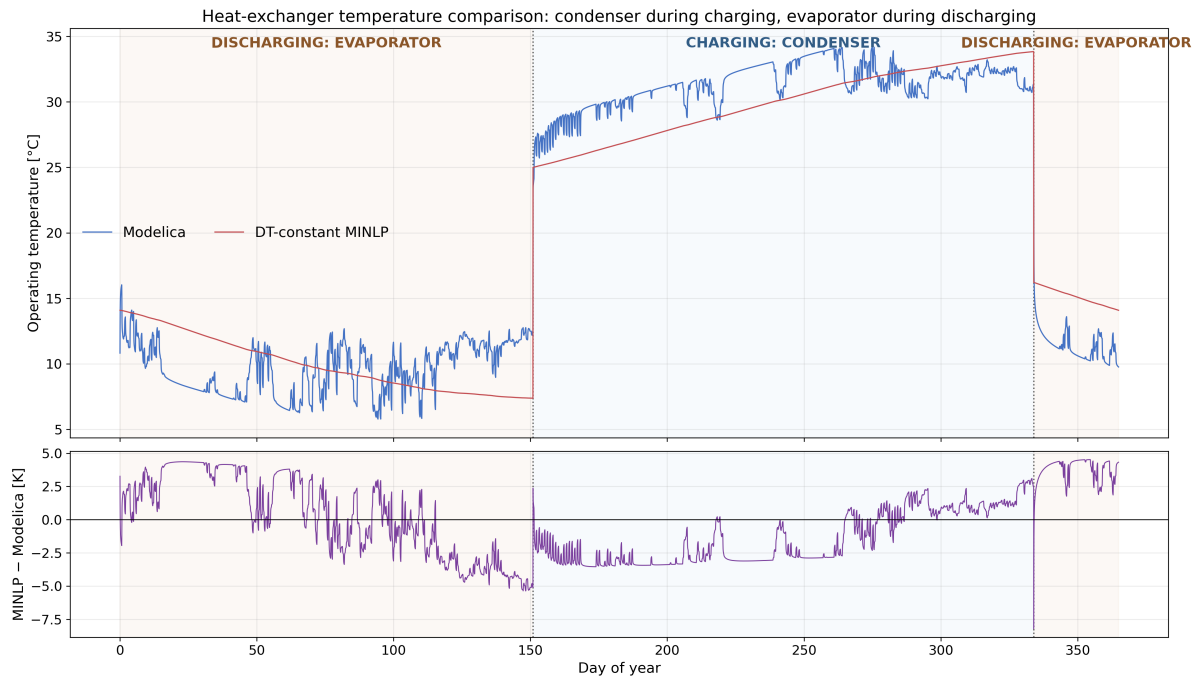


Figure 7: Operating temperatures predicted by the ΔT -constant MINLP and Modelica. The lower panel shows the paired hourly error, defined as MINLP minus Modelica. Dotted vertical lines indicate the transitions between operating periods.

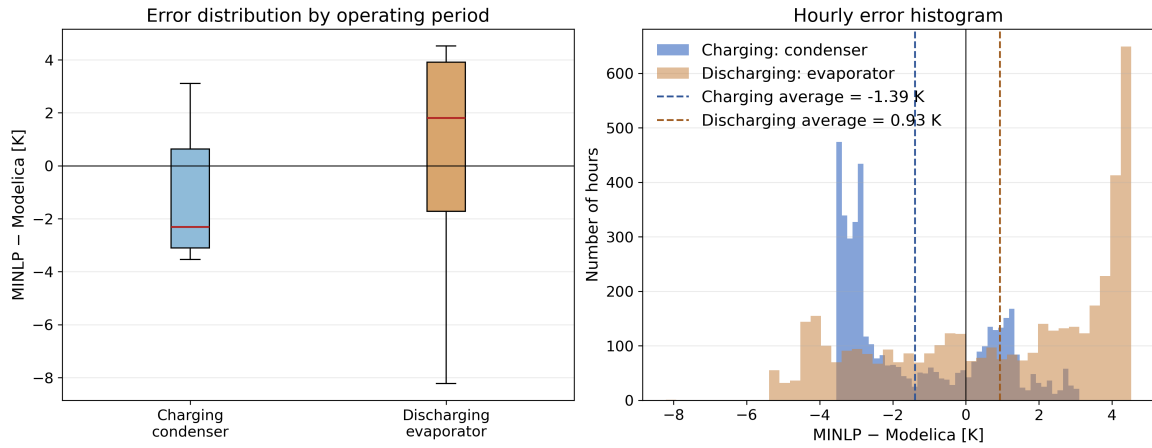


Figure 8: Distributions of the condenser-temperature error during charging and the evaporator-temperature error during discharging. The left panel presents the boxplots, and the right panel shows the hourly error histograms

3.2 Influence of Operational Strategies on BTES Sizing

Following the verification of the study, the MINLP formulation is applied to the EMPA case study shown in Figure 3 to evaluate the influence of the three operational strategies on the optimal design and operation of the integrated heating and cooling system.

Table 3 shows that the selected operational strategy affects both the technology sizing and the annual system cost. Strategy I assumes a scalar decision variable for the temperature difference (ΔT), Strategy II considers constant condenser and evaporator temperatures throughout the year, while Strategy III treats the condenser and evaporator temperatures as varying decision variables.

All three strategies select BTES design option 4, indicating that the optimal seasonal storage configuration is independent of the operational strategy for the considered case study and available options. Nevertheless, differences are observed in the sizing of the remaining technologies. Strategy II results in the highest annual cost, as fixing the condenser and evaporator temperatures prevents the optimizer from adapting the COP of the heat pump and chiller to the operating conditions. In contrast, Strategy III achieves the lowest annual cost by allowing both temperatures to vary, increasing operational flexibility and reducing the operating cost despite the slightly higher capital investment associated with larger technology capacities.

The boiler and chiller capacities remain unchanged across all strategies. As illustrated in Figure 3, the cooling demand is supplied exclusively by the BTES-connected chiller, leaving no alternative technology for cooling production. Similarly, the peak heating

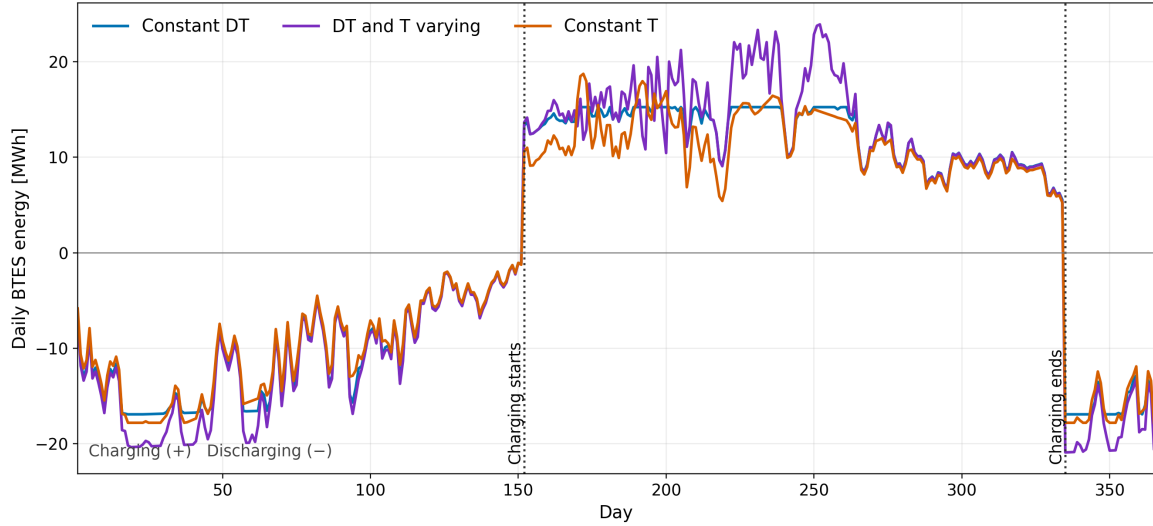
demand of 1738 kW occurs during the BTES charging period, when the boiler is the only available heating source, resulting in an identical boiler capacity for all three strategies. The solver run times are almost identical, with Strategy III converging marginally faster than the other formulations. This indicates that increasing the operational flexibility of the model does not introduce a noticeable computational penalty for the EMPA case study.

Figure 9a compares the BTES charging and discharging profiles. The discharging behavior is nearly identical for all strategies, whereas larger differences are observed during the charging period. Strategy III stores more thermal energy due to its greater operational flexibility, while the charging profiles gradually converge after approximately day 260. All three strategies identify the same charging period, suggesting that maximizing the stored thermal energy before the heating season is the preferred operating strategy for the demand profile under consideration. This behavior, however, is expected to depend on the demand characteristics and may not be universal.

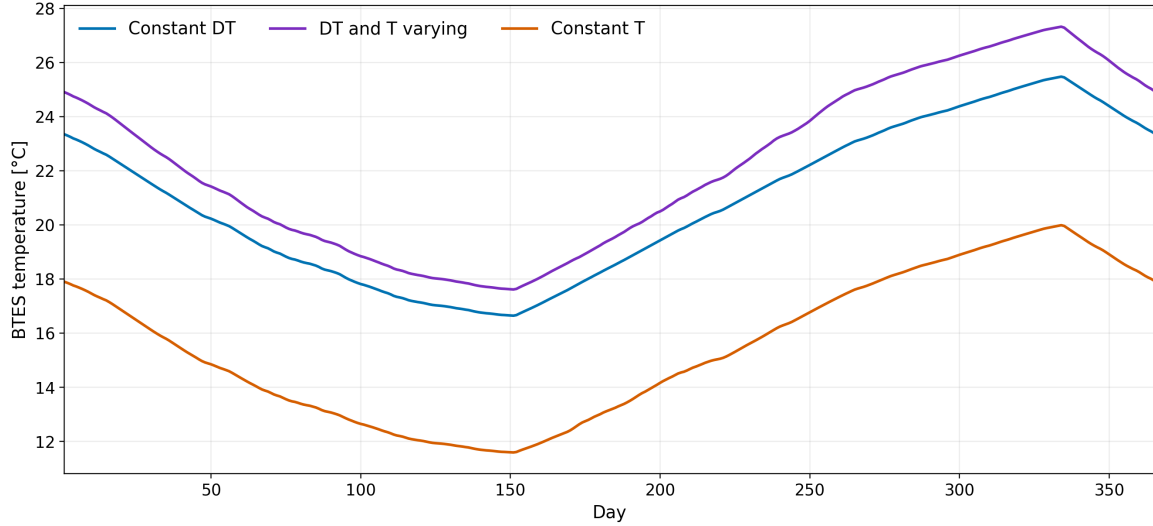
The BTES temperature profiles shown in Figure 9b are also very similar, with a maximum difference of approximately 1 °C between Strategies I and III. Strategy II maintains the lowest BTES temperatures throughout the year. Since the condenser and evaporator temperatures remain fixed, the optimizer cannot improve the COP through operational adjustments and instead relies on a lower BTES operating temperature to maximize the performance of the heat pump.

Table 3: Selected technology sizes and system costs for each modeling strategy. (Sec. 2.2)

Strategy	HP (kW)	Boiler (kW)	Chiller (kW)	Cooling Tower (kW)	BTES option	BTES Depth (m)	CAPEX (CHF/a)	OPEX (CHF/a)	Total cost (CHF/a)	Run time (s)
Strategy I										
(Constant ΔT)	1,019	1,738	1,247	810	4	80.1	194,365	793,196	987,561	42.99
Strategy II										
(Constant T)	1,139	1,738	1,247	723	4	80.1	199,122	796,168	995,291	42.63
Strategy III										
(T , ΔT vary)	1,227	1,738	1,247	610	4	80.1	202,367	766,146	968,514	37.11



(a) Daily energy supplied or extracted from the BTES during the charging and discharging period



(b) Temperature evolution of the BTES in the three different operational strategies

Figure 9: BTES performance in the three operational strategies

3.3 Assessment of Linearization Strategies

One of the main objectives of this thesis was to investigate whether the original MINLP formulation proposed by Fiorentini et al. [25] could be reformulated as a mixed-integer linear programming (MILP) model without significantly compromising the quality of the solution. The MILP formulation was developed following the methodology presented in Section 2.3. This section compares the MINLP, MILP and simulation-based heuristic models with respect to the selected technology sizes, annual system cost and computational performance. To ensure a consistent comparison, all models are applied to the EMPA case study treating the condenser and evaporator temperature as scalar decision variables as of strategy II, introduced in Section 2.2.

The initial comparison is presented in Table 4. Although the MILP is derived directly from the MINLP formulation, the two optimization models converge to noticeably different solutions. The MINLP selects BTES option 4, whereas the MILP selects the larger option 5 together with substantially smaller BTES heat pump, boiler and cooling tower capacities. These differences increase the annual system cost from CHF 995 291/a to CHF 1 056 113/a, corresponding to approximately 6.1%. This indicates that the linearization does not simply modify the numerical solution but changes the economic trade-offs sufficiently to alter the optimal system configuration.

The differences are also reflected in the computational performance. Contrary to the expectation that linear models are generally easier to solve, the MILP requires 306.98 s, compared with only 118.84 s for the MINLP. Although the bilinear constraints have been removed, the piecewise McCormick reformulation introduces six additional binary variables together with the corresponding auxiliary variables and constraints. The resulting increase in combinatorial complexity outweighs the computational benefit of removing the nonlinear terms. In contrast, the MINLP, despite containing more than 17 000 bilinear constraints, is solved efficiently by Gurobi, indicating that these nonlinear relationships are not the dominant computational bottleneck for the present case study.

An important practical advantage of the MILP formulation is its broader solver compatibility. Since all bilinear terms are eliminated, the model can be solved using a wide range of commercial and open-source MILP solvers, whereas the MINLP requires optimization software capable of handling non-convex bilinear constraints. Consequently, although the MILP is less accurate and computationally more demanding for the EMPA case study, it remains an attractive alternative when dedicated MINLP solvers are not available.

The heuristic simulation model selects the same BTES configuration as the MINLP, namely option 4, and reproduces the optimized BTES chiller capacity of 1247.12 kW. However, it predicts a BTES heat-pump capacity of 1341.90 kW, approximately 17.7% higher than the MINLP value of 1139.96 kW. The boiler capacity is also conservatively estimated at 2236.50 kW, approximately 28.6% above the optimized value of 1738.70 kW. These differences result from the peak-demand sizing rules, which provide additional capacity margins but do not account for the optimal sharing of the hourly heating demand between the available technologies.

The simulation requires only 2.51 s, making it approximately 47 times faster than the

MINLP and more than 120 times faster than the MILP. It can therefore serve as a rapid and conservative preliminary sizing method before applying a detailed optimization model.

Table 4: Technology sizes, selected BTES design, total annual cost, and runtime for the constant condenser and evaporator temperature strategy.

Quantity	MINLP	MILP	Simulation
BTES heat pump [kW]	1139.96	683.49	1341.90
Boiler [kW]	1738.70	1589.97	2236.50
BTES chiller [kW]	1247.12	1247.12	1247.12
Cooling tower [kW]	723.76	624.88	–
BTES option	4	5	4
BTES depth [m]	80.1	91.7	80.1
Total annual cost [CHF/a]	995 291	1 056 113	–
Bilinear constraints	17 520	–	–
Binary variables	26 280	26 286	–
Total runtime [s]	118.84	306.98	2.51

Rather than forcing the MILP to reproduce the MINLP solution, the following analysis investigates which modeling assumptions contribute to the observed difference. Two aspects were considered. The first is the discrete representation of the BTES design space, since the optimizer can only select from a limited number of predefined configurations. In particular, the relatively large interval between options 4 and 5 may affect the selected storage size. The second is the piecewise McCormick approximation used to replace the bilinear temperature-flow products.

Figure 10 evaluates the two and three segment McCormick formulations at the evaporator temperature selected by the MINLP, and over the heating-output range reached by the MINLP solution. The dashed line represents the exact bilinear relationship, while the shaded region contains the values permitted by the corresponding linear relaxation. The maximum envelope width decreases from 81.1 kW with two segments to 16.8 kW with three segments, confirming that the three-segment formulation provides the tighter local representation of the MINLP operating relationship.

The condenser is not included in the figure because the MINLP selects $T_{\text{cond}} = 25^\circ\text{C}$, corresponding to the lower boundary of the condenser-temperature domain. At this boundary, the McCormick constraints collapse onto the exact product for both segment choices, and no relaxation gap remains. The remaining approximation is therefore associated mainly with the evaporator term.

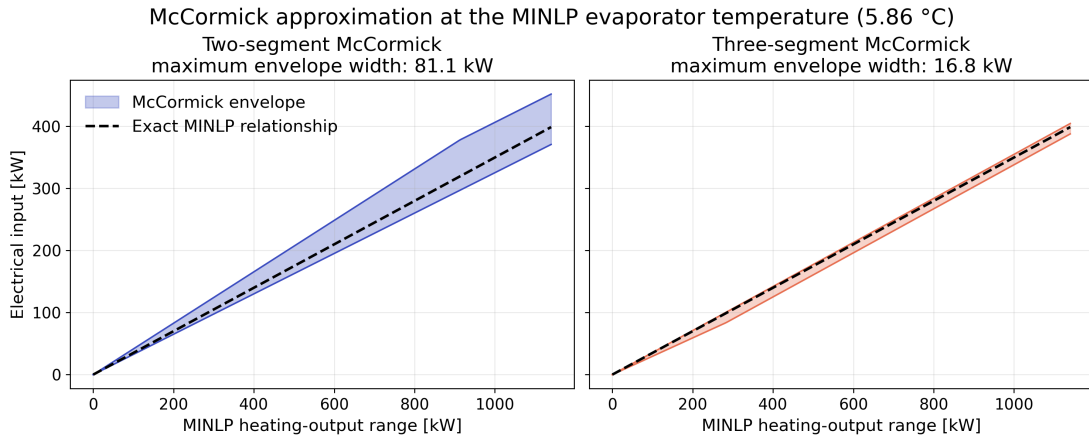


Figure 10: Two and three segment McCormick approximations of the BTES heat-pump relationship at the evaporator temperature selected by the MINLP, $T_{\text{evap}} = 5.86^\circ\text{C}$

The envelope comparison shows that the three-segment formulation represents the MINLP evaporator relationship more accurately than the two-segment formulation. However, this does not guarantee that it will reproduce the complete MINLP solution. The McCormick constraints define a feasible region rather than a unique approximation curve, and the optimizer may select any admissible point within this region. Segment boundaries also change the geometry of the feasible region and can direct the optimization toward different temperatures, equipment sizes, and operating schedules. Moreover, the BTES design remains a discrete decision. A small change in the estimated annual cost can therefore cause the optimizer to move from one storage option to another, producing a larger change in the resulting system design.

To evaluate the influence of the BTES discretization, two additional storage configurations, denoted options 4' and 4'', were introduced between the original options 4 and 5. Their parameters were obtained by linear interpolation, as summarized in Table 5.

Table 5: Original and linearly interpolated BTES design options used to refine the MILP design space.

Parameter	Option 4	Option 4'	Option 4''	Option 5
Diameter [m]	80.1	84.0	87.8	91.7
Borehole depth [m]	80.1	84.0	87.8	91.7
Volume [10^3m^3]	404.1	471.4	538.8	606.1
Thermal conductance [kW/K]	76.0	88.6	101.3	113.9
Number of boreholes	355	392	429	466
Total drilling length [m]	28 435.5	32 928.0	37 666.2	42 732.2

Introducing the intermediate configurations considerably improved the economic agreement between the two models. Instead of selecting the original option 5, the three-segment

MILP selected option 4". As shown in Table 6, the difference in annual system cost decreased from 6.11% to 0.44% relative to the MINLP. This result indicates that a significant part of the original discrepancy was caused by the coarse BTES design space.

Nevertheless, the technology capacities did not converge to the MINLP values. The refined MILP selected a larger BTES heat pump and a smaller cooling tower. The additional storage option therefore allowed the optimizer to obtain a similar annual cost through a different combination of storage and conversion equipment, rather than reproducing the MINLP design and operation directly.

A second sensitivity analysis examined the influence of the temperature segmentation. The number of McCormick segments was reduced from three to two, while the remaining modelling assumptions were maintained. The two-segment MILP selected the same BTES option as the MINLP, and its annual cost differed by only -0.77% . The BTES heat-pump capacity also moved closer to the MINLP value, while the cooling-tower capacities became nearly identical.

This result should not be interpreted as evidence that two segments provide a more accurate representation of the bilinear relationship. As demonstrated in Figure 10, the two-segment envelope is considerably wider at the MINLP evaporator temperature. Instead, changing the segment boundaries modifies the feasible relaxation and leads the optimizer to a different region of the design space. In this case, that region happens to produce technology sizes and an objective value closer to the MINLP solution. The agreement is therefore solution-specific and does not establish a general accuracy advantage for the two-segment formulation.

Some differences also remain in the refined solution. The two-segment MILP achieves a lower annual cost than the MINLP. Since the MILP is based on a relaxation of the nonlinear relationships, a lower objective value does not necessarily correspond to a physically more accurate design, but may instead reflect the additional freedom introduced by the linearized feasible region.

Table 6: Comparison of the original MINLP and MILP solutions with the interpolated BTES design options and the two-segment linearization.

Quantity	MINLP	3 seg. MILP	Int. MILP	2-seg. MILP
Linearization segments	–	3	3	2
BTES heat pump [kW]	1139.96	683.49	1220.82	1084.14
Boiler [kW]	1738.70	1589.97	1738.70	1589.97
BTES chiller [kW]	1247.12	1247.12	1247.12	1247.12
Cooling tower [kW]	723.76	624.88	610.02	725.19
BTES option	4	5	4 [*]	4
BTES depth [m]	80.1	91.7	87.8	80.1
Total annual cost [CHF/a]	995 291	1 056 113	999 631	987 675
Difference from MINLP [%]	–	6.11	0.44	-0.77
Total runtime [s]	118.84	306.98	282.80	183.76

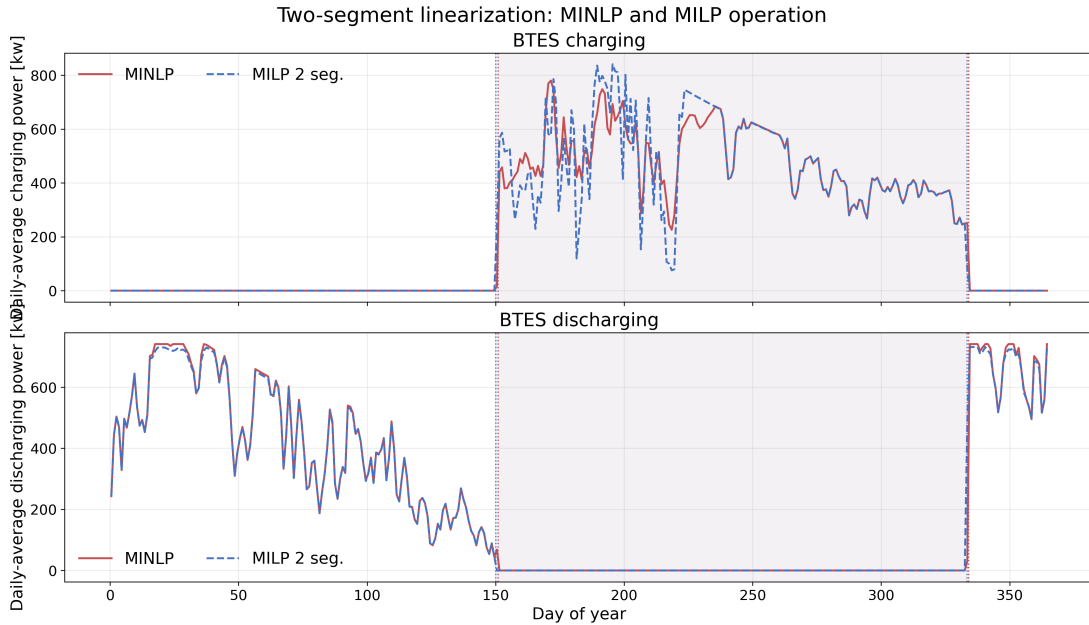


Figure 11: Daily-average BTES charging and discharging power obtained with the original MINLP and the 2 seg. MILP formulations for the EMPA constant-temperature strategy

Figure 11 illustrates the annual BTES charging and discharging profiles obtained with the MINLP and the two-segment MILP formulation. During the charging season, the MILP captures the seasonal operating pattern and follows the MINLP profile over most of the period. Some differences remain in the timing and magnitude of individual charging peaks, reflecting the flexibility available to the optimizer in scheduling rejected chiller heat between the BTES and the cooling tower. During the discharging season, the agreement is particularly close, with the two profiles almost completely overlapping. This close correspondence indicates that the two-segment linearization reproduces the principal seasonal operation of the nonlinear formulation while retaining only minor differences in the

hourly charging schedule.

Taken together, the sensitivity analyses show that the discrepancy between the MINLP and MILP formulations cannot be attributed to a single modeling assumption. Refining the BTES design space substantially reduces the economic difference, highlighting the importance of the discrete storage representation. The comparison between the two- and three-segment formulations further demonstrates that the local accuracy of the McCormick relaxation is not directly correlated with the quality of the final optimization solution. Although the two-segment formulation provides a wider relaxation envelope, it reproduces the MINLP technology sizes, annual cost, and seasonal charging and discharging profiles more closely than the three-segment formulation. This indicates that modifying the segmentation primarily changes the feasible region explored by the optimizer rather than simply improving or degrading the local approximation of the nonlinear relationship. The remaining differences are therefore best understood as the combined effect of the discrete BTES design space, the geometry of the piecewise McCormick relaxation, and the resulting annual operating strategy.

3.4 Influence of Modelling Features

An important objective of this study is to evaluate the influence of the modeling features incorporated into the MILP formulation. The analysis therefore examines how successive simplifications affect the optimal system design, the annual cost, the hourly operation, and the computational effort. This permits the trade-off between model fidelity and computational efficiency to be assessed.

The following five formulations are reported:

- **Formulation I (three-segment Empa model):** the full MILP used in the preceding Empa case-study section, with the hourly BTES charging-period binaries and three McCormick temperature segments.
- **Formulation II (one-segment model):** the formulation retains the hourly BTES charging-period binaries but uses one McCormick segment.
- **Formulation III (fixed charging period):** the charging period is fixed between days 150 and 333, eliminating the hourly binary variables that select its start, end, and active state.

- **Formulation IV (fixed period and capacities):** in addition to the fixed charging period, the boiler capacity is prescribed from the peak heating demand occurring during that period and the BTES-chiller capacity is prescribed from the peak cooling demand.
- **Formulation V (fixed temperatures):** the charging period and the two capacities remain fixed, while the evaporator and condenser temperatures are prescribed as 15 °C and 30 °C, respectively. The temperature–flow products then become linear and the McCormick construction is removed.

Table 7 summarizes the computational performance, system design, and annual cost of the different formulations. In contrast to the previous comparison, all differences are evaluated relative to the EMPA MINLP formulation, which serves as the reference model throughout this section. This allows the effect of each simplification to be assessed directly against the most accurate formulation rather than against another approximation. Since the three-segment MILP selected a different BTES design than the MINLP, the successive simplifications are evaluated in terms of how closely they reproduce the MINLP solution while reducing computational complexity.

Reducing the number of McCormick segments from three to one decreases the temperature interval resolution by a factor of three. For the BTES heat pump, the full evaporator-temperature interval of -5 to 25 °C is represented by one 30-K segment instead of three 10-K segments. For the BTES chiller, the 25 to 40 °C interval is represented by one 15-K segment instead of three 5-K segments. Because the maximum McCormick-envelope error is proportional to the width of the bounded temperature interval, the worst-case relaxation error is therefore three times larger with one segment. In a minimization problem, this enlarged feasible relaxation can produce an optimistic objective value by understating the power associated with the bilinear products. At the same time, the reduction from three segments to one removes 70 084 continuous variables and four binary variables, corresponding to 70 088 variables in total, or 19.51% of the three-segment formulation.

The one-segment formulation reports an annual system cost of 983 874 CHF/a, corresponding to a difference of only 1.15% relative to the MINLP. Furthermore, both formulations select BTES option 4, indicating that reducing the McCormick discretization from three segments to one does not alter the optimal storage size. Although the weaker relaxation increases the runtime from 118.84 s to 169.38 s, the solution remains in close

agreement with the MINLP while reducing the number of optimization variables by 70 088, corresponding to 19.51% of the original formulation.

Relative to the MINLP, the fixed-period formulation contains 26 277 fewer binary variables, reducing the number of binary variables from 26 280 to only three. Its total runtime is 90.03 s, corresponding to a 1.32-fold speedup relative to the MINLP. Its annual cost differs by only 0.88% from the MINLP, while the selected BTES design remains identical (option 4). Within the one-segment formulations, fixing the charging period changes the annual cost by only 2 652 CHF/a, indicating that the prescribed charging interval closely matches the period selected by the optimization. Nevertheless, this observation remains case-specific, since prescribing a charging period without prior knowledge of the optimal solution could lead to a suboptimal design.

Prescribing the boiler and BTES-chiller capacities produces the same annual cost and essentially the same design and operation as the fixed-period model. Its total runtime is 106.99 s, corresponding to a 1.11-fold speedup relative to the MINLP, although it is slightly slower than fixing the charging period alone. This conclusion remains case-specific: with a different portfolio or demand profile, fixing these capacities could exclude an economically preferable allocation between competing technologies.

The largest computational improvement is obtained by fixing the evaporator and condenser temperatures. The bilinear products then reduce to

$$Q_t T_{\text{evap}} = 15Q_t, \quad Q_t T_{\text{cond}} = 30Q_t, \quad (56)$$

and the McCormick auxiliary variables and constraints become unnecessary. The number of continuous variables decreases from 332 895 in the MINLP to 210 250, a reduction of 36.84%, while the number of binary variables decreases from 26 280 to zero. The total runtime falls from 118.84 s to 11.14 s, corresponding to a 10.67-fold speedup. Despite this substantial computational improvement, the formulation deviates considerably from the MINLP solution. The selected BTES design changes from option 4 to option 5, resulting in an annual system cost that is 6.23% higher than the MINLP. This indicates that prescribing the evaporator and condenser temperatures significantly alters the thermodynamic behavior of the system, affecting both the optimal system design and its hourly operation.

Figure 12 compares the operational behaviour of the three-segment MILP and four MILP

simplifications relative to the EMPA MINLP. The one-segment formulation exhibits the closest agreement with the reference solution, with a BTES temperature NRMSE of only 1.3%. Fixing the charging period produces a similarly small temperature deviation of 1.6%, while additionally prescribing the boiler and BTES-chiller capacities introduces no noticeable further change, maintaining a temperature NRMSE of approximately 1.6%. In contrast, the three-segment and fixed-temperature formulations exhibit substantially larger temperature deviations, with NRMSE values of 44.9% and 44.2%, respectively.

The hourly charging profiles show considerably larger deviations. The one-segment, fixed-period, and fixed-period/capacity formulations exhibit charging NRMSE values of 39.8%, 39.2%, and 39.7%, respectively. Despite these differences in hourly operation, their aggregated annual charging energies differ from the MINLP by only 4.2%, 5.0%, and 5.0%. Their discharging profiles remain closer to the reference, with NRMSE values of 12.5%, 8.9%, and 8.9%, respectively. For comparison, the three-segment and fixed-temperature formulations have charging NRMSE values of 41.3% and 47.4%, aggregated charging-energy differences of 5.9% and 6.1%, and discharging NRMSE values of 27.5% and 26.6%, respectively.

The combination of relatively small temperature and discharging deviations with larger hourly charging deviations is physically explainable. BTES temperature is a state variable governed by the cumulative net-energy balance of a large thermal mass. Differences in hourly charging are therefore integrated over time and smoothed by the storage heat capacity, allowing different charging schedules to produce similar temperature trajectories. This interpretation is supported by the aggregated charging-energy results: although the hourly charging NRMSE is approximately 39%–47%, the total annual charging energy differs from the MINLP by only 4%–6%. Charging also has considerable scheduling flexibility because rejected chiller heat can be directed either to the BTES or to the cooling tower. Consequently, shifting intermittent charging peaks between hours can produce a relatively large hourly profile NRMSE even when a similar total amount of energy is supplied to the storage over the year. In contrast, discharging is more constrained because it is directly coupled to the hourly heating demand and competes with the boiler. Its timing therefore, follows the heating demand more closely. Thus, a close temperature trajectory, a relatively small difference in the discharging profile, and a larger difference in the hourly charging profile are not contradictory. The close temperature trajectories of

the one-segment, fixed-period, and fixed-period/capacity formulations are consistent with their selection of the same BTES option as the MINLP and their ability to reproduce its overall storage behavior despite differences in hourly charging schedules. By contrast, the three-segment and fixed-temperature formulations select a different BTES option and exhibit larger operational deviations. Prescribing the evaporator and condenser temperatures additionally changes the thermodynamic operating conditions throughout the year, contributing to the deviations observed for the fixed-temperature formulation.

Overall, the successive simplifications reduce computational effort relative to the original three-segment MILP. Among them, the one-segment formulation provides the best balance between computational efficiency and solution accuracy, reproducing the MINLP system design while reducing the annual system cost by only 1.15%. Fixing the charging period provides an additional reduction in computational effort with only a marginal change in the solution, while prescribing the boiler and BTES-chiller capacities introduces virtually no further differences. Removing the McCormick representation entirely yields the largest computational improvement, reducing the runtime by more than one order of magnitude. However, this simplification also produces the largest deviations in both system design and operation, demonstrating that simplifying the temperature-dependent thermodynamic relationships has a substantially greater impact on solution quality than simplifying the charging schedule or fixing the equipment capacities.

Table 7: Comparison of the formulations for the constant condenser and evaporator temperature strategy.

Quantity	MINLP	3 seg.	1 seg.	Fixed period	Fixed period/sizes	Fixed temperatures
McCormick temperature segments	–	3	1	1	1	0
Continuous variables	332 895	332 895	262 812	262 812	262 812	210 250
Binary variables	26 280	26 286	26 283	3	3	0
Total runtime [s]	118.84	306.98	169.38	90.03	106.99	11.14
Speedup relative to MINLP	1.00	0.40	0.70	1.32	1.11	10.67
BTES heat pump [kW]	1139.96	683.49	956.72	951.33	951.33	688.43
Boiler [kW]	1738.70	1589.97	1589.97	1738.70	1738.70	1738.70
BTES chiller [kW]	1247.12	1247.12	1247.12	1247.12	1247.12	1247.12
Cooling tower [kW]	723.76	624.88	725.74	723.66	723.66	624.88
BTES option	4	5	4	4	4	5
BTES depth [m]	80.1	91.7	80.1	80.1	80.1	91.7
CAPEX [CHF/a]	199 122	220 051	190 332	191 127	191 127	221 293
OPEX [CHF/a]	796 169	836 062	793 543	795 398	795 398	836 007
Total annual cost [CHF/a]	995 291	1 056 113	983 874	986 526	986 526	1 057 300
Cost difference from MINLP [%]	0.00	6.11	-1.15	-0.88	-0.88	6.23
Temperature RMSE [K]	0.000	7.064	0.200	0.248	0.252	6.957

All differences, speedups, and RMSE values use the EMPA MINLP as the reference.

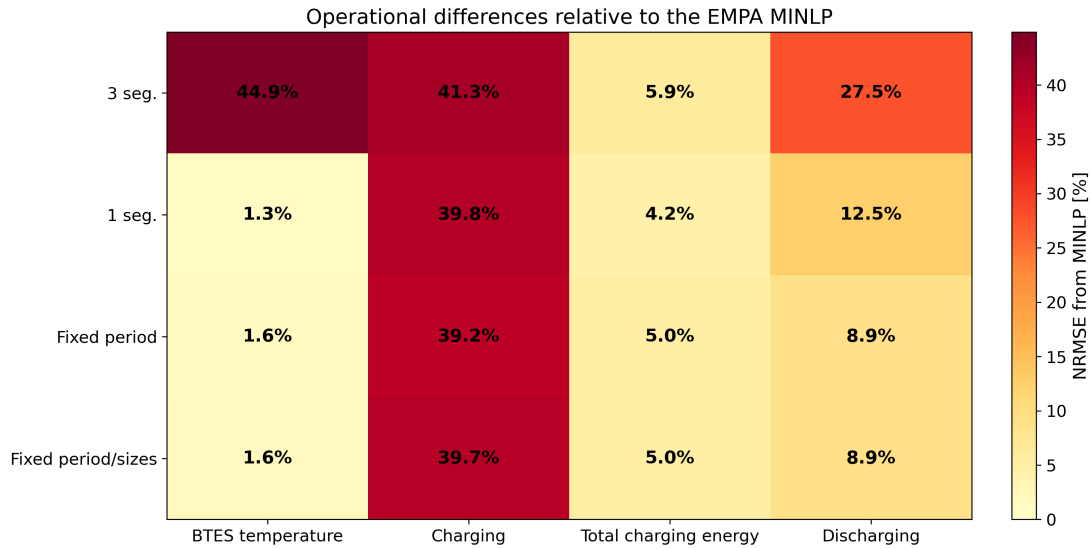


Figure 12: Normalized root-mean-square differences of all four linearization formulations relative to EMPA MINLP

The statistical procedure is documented in Appendix A3.

3.5 Influence of System Complexity

The purpose of this section is to investigate how the increased complexity of the extended case study affects the computational performance of the proposed optimization formulations. Compared with the simpler EMPA configuration, the extended system contains additional air-source technologies, solar thermal collectors, and short-term thermal storage. As expected, the additional technologies and operational interactions increase the computational effort required by the extended-system MINLP.

Based on the modeling-feature analysis in the preceding section, the one-segment MILP with a fixed BTES charging period is considered here. This formulation showed a promising compromise between model fidelity and computational efficiency.

For the extended case study, the MILP performs better computationally than the MINLP. Its total runtime decreases from 366.73 to 253.37 s, corresponding to a reduction of 30.91% and a 1.45-fold speedup. This result supports the conclusion of the preceding section that the one-segment, fixed-period formulation becomes particularly useful when the complexity of the overall energy system increases.

The two formulations produce broadly consistent system designs. Both select BTES option 1 and give identical BTES-chiller and air-chiller capacities. The largest capacity difference occurs for the BTES heat pump, which is 10.11% larger in the MILP, while the remaining technology capacities differ by approximately 2% or less. The MILP reports a

total annual cost 2.80% below the MINLP and a BTES-temperature RMSE of 0.808 K. The lower objective reflects the effect of the linear approximation and should not be interpreted as evidence of a physically superior design. Overall, the one-segment, fixed-period MILP reduces the computational burden while retaining the main design and operational characteristics of the extended-system MINLP. The MILP should therefore be interpreted as a computationally efficient approximation of the MINLP.

Table 8: Comparison of the extended case study MINLP and one-segment, fixed-period MILP for the constant condenser and evaporator temperature strategy.

Quantity	Unit	Ext. MINLP	Ext./1 seg./Fixed period MILP	MILP-MINLP [%]
Total runtime	s	366.73	253.37	-30.91
Speedup relative to MINLP	–	1.00	1.45	–
BTES heat pump	kW	296.87	326.89	+10.11
Air-source heat pump	kW	1 064.34	1 043.43	-1.97
Boiler	kW	875.28	866.19	-1.04
BTES chiller	kW	890.30	890.30	0.00
Air chiller	kW	356.82	356.82	0.00
Cooling tower	kW	829.53	830.04	+0.06
Solar thermal area	m ²	1 279.48	1 263.85	-1.22
Short-term thermal storage	kWh	398.35	393.49	-1.22
BTES option	–	1	1	0.00
BTES depth	m	53.4	53.4	0.00
CAPEX	CHF/a	191 932	191 678	-0.13
OPEX	CHF/a	633 138	610 322	-3.60
Total annual cost	CHF/a	825 069	802 000	-2.80
BTES-temperature RMSE	K	–	0.808	–

4 Conclusions

This thesis investigated how different levels of modeling fidelity influence the design, operation, and computational performance of integrated heating and cooling systems incorporating Borehole Thermal Energy Storage (BTES) system. Starting from the nonlinear optimization framework proposed by Fiorentini et al., the work systematically examined three complementary dimensions of model development: the thermodynamic assumptions governing BTES operation, the mathematical formulation of the optimization problem, and the influence of modeling simplifications and system complexity. Rather than seeking the most accurate or the fastest model alone, the objective was to identify which modeling features are essential for reliable design decisions and which can be simplified without significantly affecting the resulting solution.

The optimization framework was first verified against a detailed Modelica simulation. Although the simplified optimization model cannot reproduce transient heat-transfer effects or partial-load operation, it accurately captured the seasonal evolution of the BTES operating temperatures, with an annual mean absolute error of 2.51 K and a root-mean-square error of 2.84 K. This verification demonstrated that the adopted BTES representation provides sufficient accuracy for annual techno-economic design optimization while maintaining a computational complexity that is suitable for optimization.

With respect to the first objective of this thesis, the comparison of the three operational strategies showed that the operational assumptions used to represent the interaction between the BTES, the heat pump, and the chiller directly influence the optimal system design. Allowing the evaporator and condenser temperatures to vary throughout the year consistently produced the lowest annual system cost without any increase in computational effort. The results therefore indicate that increasing operational flexibility provides tangible economic benefits without significantly compromising computational tractability. Regarding the influence of the mathematical formulation, the comparison between the MINLP and MILP formulations demonstrated that replacing nonlinear equations with linear approximations does not automatically improve computational performance. For the investigated EMPA case study, the original three-segment McCormick reformulation introduced additional binary variables and combinatorial complexity that outweighed the benefits of removing the bilinear terms, resulting in longer solution times than the MINLP. However, the subsequent analyses showed that these differences cannot be attributed

solely to the linearization itself. Refining the BTES design space and reducing the McCormick discretization to a single segment produced solutions that closely reproduced the MINLP while requiring substantially lower computational effort than the original three-segment MILP. This demonstrates that the practical performance of the MILP depends not only on the mathematical approximation itself but also on how the linearization interacts with the discrete design space and the operating conditions of the system.

A detailed assessment of the McCormick approximation provided one of the principal methodological findings of this work. Although increasing the number of temperature segments generally tightens the relaxation over the complete operating domain, the results demonstrated that this does not necessarily translate into a solution that is closer to the original MINLP. In both case studies, the one-segment MILP reproduced the nonlinear solution remarkably well despite its simpler approximation. This behavior is explained by the fact that the local approximation accuracy depends on the position of the optimal operating point within the McCormick partition rather than on the total number of segments alone. Consequently, increasing the number of segments increases computational complexity but does not necessarily improve the quality of the final design decisions.

The systematic simplification study further showed that different modeling assumptions influence the optimization problem to very different degrees. The one-segment MILP maintained excellent agreement with the MINLP while substantially reducing the computational effort compared with the original three-segment formulation. Furthermore, when the seasonal charging period is known, fixing this period provides an additional reduction in computational effort with only a negligible influence on the optimal solution. Prescribing the boiler and BTES-chiller capacities introduced virtually no further changes, indicating that these capacities are largely dictated by the peak thermal demands. In contrast, prescribing the evaporator and condenser temperatures produced the largest computational improvement but also the largest deviation from the nonlinear reference solution. These results suggest that, for applications where the charging period can be estimated reliably, the one-segment fixed-period MILP provides the best compromise between computational efficiency and solution accuracy.

The steady-state heuristic formulation demonstrated that simple engineering calculations based on aggregated seasonal energy balances can provide very rapid preliminary sizing of BTES systems. The heuristic cannot reproduce the operational flexibility or globally

optimal solutions obtained with mathematical optimization, but it can be used in early-stage feasibility studies, where many design alternatives must be screened before detailed optimization is performed.

The application of the methodology to the extended integrated energy system provided an additional important insight into the scalability of the proposed formulations. Although the larger system increased the computational effort of the MINLP formulation, the one-segment fixed-period MILP solved the extended case study with a higher speedup compared to the MINLP than the corresponding EMPA case study while maintaining close agreement with the nonlinear reference solution. This result demonstrates that computational effort is governed not only by the size of the optimization problem but also by the degree of operational flexibility available within the energy system. By introducing additional heating and cooling technologies, the optimization can satisfy the demand through a wider range of feasible technology combinations, reducing its dependence on the nonlinear BTES subsystem. Consequently, the computational advantage of the simplified MILP formulation becomes more pronounced as system flexibility increases.

Overall, this thesis demonstrates that there is no universally optimal modeling approach for BTES-integrated energy systems. Instead, the appropriate formulation depends on the intended balance between computational effort and solution accuracy. The MINLP remains the preferred formulation when the highest level of modeling fidelity is required, as it provides the most accurate representation of the thermodynamic interactions governing the system. However, the results of both case studies showed that the one-segment MILP is capable of reproducing the MINLP solution with only minor differences while avoiding the computational burden associated with the original three-segment formulation. Furthermore, when the seasonal charging period is known, the one-segment fixed-period MILP provides an even more computationally efficient alternative, making it particularly attractive for the optimization of larger and more flexible integrated energy systems.

Appendix

A1 Input Data

The main input parameters used in the mathematical models are summarized in Tables A1–A4. Only parameters appearing directly in the model formulation are reported.

Table A1: General model parameters

Symbol	Value	Unit	Description
i	0.04	–	Interest rate
Δt	1	h	Simulation timestep
T_{ground}	12	°C	Undisturbed ground temperature

Table A2: BTES thermal parameters

Symbol	Value	Unit	Description
c_v	2200	$\text{kJ m}^{-3} \text{K}^{-1}$	Volumetric heat capacity of the ground
k_g	2.4	$\text{W m}^{-1} \text{K}^{-1}$	Ground thermal conductivity
U_{top}	0.14	$\text{W m}^{-2} \text{K}^{-1}$	Top insulation heat transfer coefficient
hlf	21.2	–	Heat loss factor
ΔT^{min}	5	K	Minimum heat-exchanger temperature difference
ΔT^{max}	15	K	Maximum heat-exchanger temperature difference

Table A3: BTES-connected heat pump and chiller parameters

Symbol	Value	Unit	Description
a_{hp}	-0.00592	$^{\circ}\text{C}^{-1}$	Heat pump inverse-COP coefficient
b_{hp}	0.3844	–	Heat pump inverse-COP coefficient
a_{ch}	0.00552	$^{\circ}\text{C}^{-1}$	Chiller inverse-COP coefficient
b_{ch}	-0.00514	–	Chiller inverse-COP coefficient
$T_{\text{evap}}^{\text{min}}$	-5	°C	Minimum evaporator temperature
$T_{\text{evap}}^{\text{max}}$	25	°C	Maximum evaporator temperature
$T_{\text{cond}}^{\text{min}}$	25	°C	Minimum condenser temperature
$T_{\text{cond}}^{\text{max}}$	40	°C	Maximum condenser temperature
$S_{\text{hp,BTES}}^{\text{max}}$	3307	kW	Maximum BTES heat pump capacity
$S_{\text{ch,BTES}}^{\text{max}}$	5000	kW	Maximum BTES chiller capacity
COP_{hp}	3.8	–	Constant heat pump COP used in the heuristic model
COP_{ch}	5.9	–	Constant chiller COP used in the heuristic model

Table A4: Technology and economic parameters

Symbol	Value	Unit	Description
η_{boiler}	0.92	–	Boiler efficiency
$S_{\text{boiler}}^{\text{max}}$	5000	kW	Maximum boiler capacity
$S_{\text{CT}}^{\text{max}}$	13900	kW	Maximum cooling tower capacity
UA_{CT}	10000	kW K ⁻¹	Cooling tower heat-transfer conductance
η_{sol}	0.65	–	Solar thermal collector efficiency
$A_{\text{sol}}^{\text{max}}$	10000	m ²	Maximum solar collector area
λ_{hp}	576	CHF kW ⁻¹	Heat pump specific investment cost
λ_{ch}	576	CHF kW ⁻¹	Chiller specific investment cost
λ_{boiler}	120	CHF kW ⁻¹	Boiler specific investment cost
λ_{CT}	55	CHF kW ⁻¹	Cooling tower specific investment cost
λ_{sol}	500	CHF m ⁻²	Solar thermal specific investment cost
λ_{STTS}	9	CHF kWh ⁻¹	Short-term thermal storage specific investment cost
n_{BTES}	60	yr	BTES economic lifetime
λ_{GHX}	66	CHF m ⁻¹	Specific drilling cost of the ground heat exchanger

The predefined BTES design options used throughout this work are listed in table 1 and are adopted unchanged for all optimization models.

A2 Mathematical Formulation of the Steady-State Heuristic Model

A2.1 Seasonal Energy Assessment

The hourly heating and cooling demand profiles are aggregated into seasonal energy quantities,

$$E_{\text{heat}} = \sum_{t \in \mathcal{T}} Q_{\text{dem,h},t} \Delta t, \quad (\text{A1})$$

$$E_{\text{cool}} = \sum_{t \in \mathcal{T}} Q_{\text{dem,c},t} \Delta t. \quad (\text{A2})$$

Assuming a constant chiller coefficient of performance, the rejected condenser heat available for storage is

$$E_{\text{rej}} = E_{\text{cool}} \left(1 + \frac{1}{\text{COP}_{\text{ch}}} \right). \quad (\text{A3})$$

Similarly, the seasonal heat extracted from the BTES is estimated from the heat pump energy balance,

$$E_{\text{BTES,req}} = E_{\text{heat}} \left(1 - \frac{1}{COP_{\text{hp}}} \right). \quad (\text{A4})$$

The seasonal storage target is therefore

$$E_{\text{BTES,target}} = \min (E_{\text{rej}}, E_{\text{BTES,req}}). \quad (\text{A5})$$

A2.2 Equipment Sizing

The charging chiller and the boiler capacity are assumed equal to the peak cooling demand and heating demand respectively,

$$S_{\text{ch,BTES}} = Q_{\text{cool}}^{\text{max}}, \quad (\text{A6})$$

$$S_{\text{boiler}} = Q_{\text{heat}}^{\text{max}}, \quad (\text{A7})$$

whereas the BTES heat pump is sized as

$$S_{\text{hp,BTES}} = 0.6Q_{\text{heat}}^{\text{max}}. \quad (\text{A8})$$

A2.3 BTES Design Selection

For each candidate BTES design, the maximum storable thermal energy is estimated as

$$E_{\text{BTES,max}} = C_{\text{BTES}} (T_{\text{BTES}}^{\text{max}} - T_{\text{BTES}}^{\text{min}}), \quad (\text{A9})$$

where C_{BTES} is the thermal capacity of the storage.

The smallest BTES configuration satisfying

$$E_{\text{BTES,max}} \geq E_{\text{BTES,target}} \quad (\text{A10})$$

is selected for the subsequent simulation.

A2.4 Daily Storage Simulation

The selected BTES is simulated using a lumped daily energy balance. During the charging season, thermal energy supplied by the chiller condenser is added to the storage, while during winter thermal energy is extracted by the heat pump.

At each daily time step, thermal losses through the top insulation and the surrounding ground are evaluated and subtracted from the stored energy. The storage temperature is updated according to the energy balance presented in Eq. 18.

The storage temperature is constrained between the prescribed minimum and maximum operating temperatures. Thermal energy exceeding the upper temperature limit is considered curtailed charging energy, whereas insufficient stored energy during discharge is recorded as unmet heating demand.

The resulting annual temperature profile is used to evaluate the seasonal performance of the selected BTES configuration.

A3 Evaluation Metrics for the MILP Simplifications

The successive MILP simplifications were evaluated against the MINLP formulation to quantify their effects on temporal accuracy, annual system cost, and computational performance. The MINLP was selected as the reference because it is the most detailed model.

Root-Mean-Square Error For every model, the hourly results were aligned by timestep over the complete annual horizon of $N = 8760$ hours. For a time-dependent quantity x , the error at timestep t was defined as

$$e_t = x_t^{\text{simplified}} - x_t^{\text{MINLP}}, \quad (\text{A11})$$

where x_t^{MINLP} is the value obtained from the MINLP and $x_t^{\text{simplified}}$ is the corresponding value from a simplified, MILP formulation.

The root-mean-square error (RMSE) [38] was calculated as

$$\text{RMSE}(x) = \sqrt{\frac{1}{N} \sum_{t=1}^N \left(x_t^{\text{simplified}} - x_t^{\text{MINLP}} \right)^2}. \quad (\text{A12})$$

Squaring the hourly errors gives greater weight to large deviations, making the metric sensitive to differences in peak operation and operational timing.

Normalized Root-Mean-Square Error Because temperature and thermal-power profiles have different units and numerical scales, a normalized RMSE was also calculated. The normalization was based on the root-mean-square value of the MINLP model reference profile:

$$x_{\text{RMS}}^{\text{MINLP}} = \sqrt{\frac{1}{N} \sum_{t=1}^N (x_t^{\text{MINLP}})^2}. \quad (\text{A13})$$

The normalized root-mean-square error (NRMSE) was then calculated as

$$\text{NRMSE}(x) = 100 \frac{\text{RMSE}(x)}{x_{\text{RMS}}^{\text{MINLP}}}. \quad (\text{A14})$$

Increasing values indicate progressively greater temporal deviation from the reference solution.

Normalization by the reference RMS was selected instead of normalization by the annual mean because the BTES charging and discharging profiles contain many zero-valued timesteps. Their mean values can consequently be small, producing excessively large or unstable normalized errors. RMS normalization instead provides a scale based on the magnitude of the complete reference profile.

Annual-Cost Difference The economic effect of each simplification was evaluated using the percentage difference in total annual cost relative to the MINLP:

$$\Delta J_m = 100 \frac{J_{\text{simp}}^{\text{total}} - J_{\text{MINLP}}^{\text{total}}}{|J_{\text{MINLP}}^{\text{total}}|}, \quad (\text{A15})$$

where $J_{\text{simp}}^{\text{total}}$ is the total annual cost of simplified MILP model m , and $J_{\text{MINLP}}^{\text{total}}$ is the total annual cost obtained from the MINLP.

A lower cost from a simplified model does not necessarily represent a physically superior design. It may result from approximations or restrictions that modify the feasible region or operational representation.

Computational Speedup The computational benefit of each simplification was quantified using its total-runtime speedup relative to the MINLP:

$$S_m = \frac{t_{\text{MINLP}}^{\text{total}}}{t_m^{\text{total}}}, \quad (\text{A16})$$

Total runtime includes model construction, optimization, and result extraction.

The speedup is interpreted as

$$\begin{cases} S_m > 1, & \text{the simplified model is faster,} \\ S_m = 1, & \text{the models require the same runtime,} \\ S_m < 1, & \text{the simplified model is slower.} \end{cases} \quad (\text{A17})$$

For example, a speedup of 48.14 indicates that the simplified model is approximately 48 times faster than the full MILP. Conversely, a speedup of 0.98 indicates that the simplified formulation is slightly slower.

Runtime depends on the hardware, solver version, solver settings, termination criteria, and numerical characteristics of each formulation. Consequently, the reported speedups should be interpreted as case-specific computational indicators rather than universally reproducible properties.

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